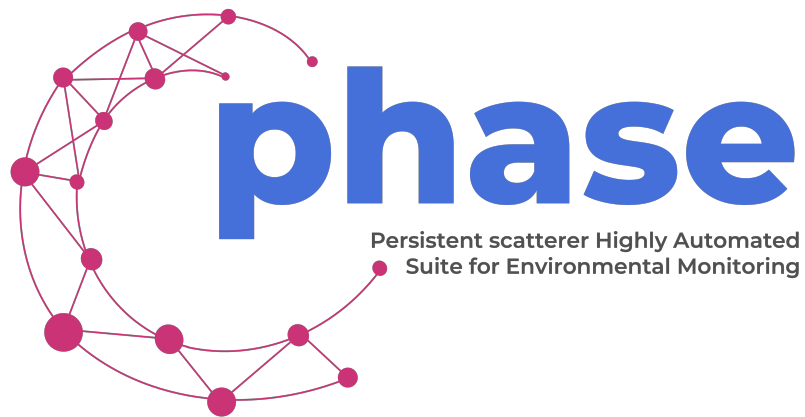
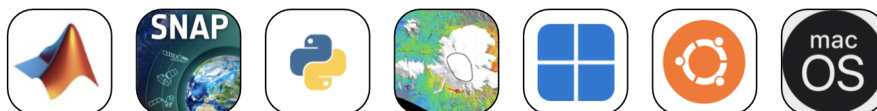




USER MANUAL

Version 4.0



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Last update: April 10, 2026

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Introduction

The PHASE software suite is an end-to-end processing chain designed to accelerate, automate, and analyze Synthetic Aperture Radar (SAR) interferometric products. Built explicitly around the Sentinel-1 constellation and its open-access data policy, the suite streamlines the established SNAP-StaMPS workflow while seamlessly supporting COSMO-SkyMed (CSK/CSG) products through a unified preprocessing pipeline. The architecture is composed of two primary components: (i) a preprocessing module that handles the ingestion, coregistration, interferogram generation of raw SAR data, and the PSI technique applied through StaMPS, and (ii) an advanced spatio-temporal modeling module dedicated to the analysis of Persistent Scatterer (PS) time series. Together, these tools provide a comprehensive framework capable of guiding the user from raw data processing through to the physical interpretation of ground deformation phenomena.

Updates history

The software has evolved significantly to minimize manual configuration and maximize user-friendliness.

v3.0

1. Introduction of the dedicated spatio-temporal modeling component (Module 2).

v4.0

Version 4.0 focuses on comprehensive automation. Tedious manual configurations have been completely eliminated:

1. Users now define their Area of Interest (AOI) interactively through a graphical map interface for both Sentinel-1 and COSMO-SkyMed.

2. Under the hood, the software automatically manages orbit file retrieval, temporal bounds, and the precise identification of Sentinel-1 swaths and bursts.
3. Introduction of an intelligent master image selection tool that queries meteorological APIs to identify the optimal master image for the stack—favoring end-of-summer dates with low moisture and no precipitation, while maintaining temporal centrality in the time series.
4. Dropped support for the deprecated Python 2.

This manual serves as a comprehensive guide to all processing steps and configuration options within PHASE. To provide a centralized reference and save users from constantly switching between different documents, relevant instructions from third-party software manuals (such as SNAP and StaMPS) have been integrated directly into this text where necessary.

System Requirements & Compatibility

The software relies on modern libraries and requires a properly configured **Python 3** environment. Additionally, the installation of **SNAP 9.x is strictly mandatory** for the preprocessing pipeline to function correctly. The suite has been developed and rigorously tested on Windows, which remains the guaranteed platform for full stability and functionality. However, the architecture is fully designed to support execution on Linux and macOS environments as well.

Provided software components

The distribution package contains the following core components:

- `PHASE_Preprocessing.mlapp`
- `PHASE_Preprocessing` folder:
 - `PHASE_StaMPS.mlapp`
 - `snap2stamps` folder
- `PHASE_model.mlapp`
- `Extra` folder:
 - `NaturalEarth` folder
- `geoSplinter` folder:
 - `linux` folder
 - `macos_apple_silicon` folder
 - `macos_intel` folder
 - `windows` folder

- MatlabFunctions folder
- pythonScripts folder
- PHASE_logo.png
- PHASE_mod1a.png
- PHASE_mod1b.png
- PHASE_mod2.png

Nome	Data di modifica	Dimensioni	Tipo
Extra	oggi, 14:45	--	Cartella
> NaturalEarth	7 apr 2026, 12:26	--	Cartella
geoSplinter	7 apr 2026, 12:26	--	Cartella
> linux	7 apr 2026, 12:26	--	Cartella
> macos_apple_silicon	7 apr 2026, 12:26	--	Cartella
> macos_intel	7 apr 2026, 12:26	--	Cartella
> windows	7 apr 2026, 12:26	--	Cartella
MatlabFunctions	7 apr 2026, 12:26	--	Cartella
PHASE_logo.png	9 set 2025, 09:47	545 KB	Immagine PNG
PHASE_Manual.pdf	26 mar 2026, 15:32	14,7 MB	PDF
PHASE_mod1a.png	9 set 2025, 10:04	571 KB	Immagine PNG
PHASE_mod1b.png	9 set 2025, 10:58	447 KB	Immagine PNG
PHASE_mod2.png	9 set 2025, 14:40	547 KB	Immagine PNG
PHASE_model.mlapp	27 gen 2026, 16:22	240 KB	MATLAB App
> PHASE_Preprocessing	7 apr 2026, 17:07	--	Cartella
> PHASE_StaMPS.mlapp	7 apr 2026, 17:07	391 KB	MATLAB App
> snap2stamps	7 apr 2026, 12:26	--	Cartella
> PHASE_Preprocessing.mlapp	oggi, 14:15	413 KB	MATLAB App
> pythonScripts	7 apr 2026, 12:26	--	Cartella

Figure 1.1: Folder/subfolder structure of the shared PHASE code.

Tested environments

The entire procedure was validated on the following system configurations:

- **SNAP:** version 9.x (Mandatory)
- **Python:** version 3.11
- **Operating systems:** Windows 10 (Guaranteed), Ubuntu 20.04, and macOS Sequoia (15.1)
- **MATLAB:** R2025a

References and acknowledgments

When utilizing PHASE Module 1 for your research, please ensure you refer to and cite the foundational works that make this pipeline possible:

- R. Monti and L. Rossi. PHASE: a Matlab-based software for the DInSAR PS processing. *Geodesy and Cartography*, 51(2):88–99, 2025. doi: 10.3846/gac.2025.21995

- M. Fomelis, J. M. Delgado Blasco, Y. L. Desnos, M. Engdahl, D. Fernández, L. Veci, J. Lu, and C. Wong. SNAP-StaMPS integrated processing for Sentinel-1 Persistent Scatterer Interferometry. In *IGARSS 2018 - 2018 IEEE International Geoscience and Remote Sensing Symposium*, pages 1364–1367. IEEE, 2018. doi: 10.1109/IGARSS.2018.8519545
- A. Hooper. A multi-temporal InSAR method incorporating both persistent scatterer and small baseline approaches. *Geophysical Research Letters*, 35 (16):L16302, 2008. doi: 10.1029/2008GL03465
- (if TRAIN is used) D. P. S. Bekaert, R. J. Walters, T. J. Wright, A. J. Hooper, and D. J. Parker. Statistical comparison of InSAR tropospheric correction techniques. *Remote Sensing of Environment*, 170:40–47, 2015. doi: 10.1016/j.rse.2015.08.035

The Authors express their gratitude to Jose Manuel Delgado Blasco and Dr. Michael Fomelis for the development of the snap2stamps tool and Prof. Andy Hooper for the development of StaMPS.

Module 1

SAR images preprocessing



2.1 Preliminary steps

This module of PHASE requires the following MATLAB packages to be installed:

- Mapping Toolbox.

2.1.1 SNAP Installation and Configuration

The first step is to download and install the SNAP software from the European Space Agency website, as the processing relies heavily on its libraries.

REQUIREMENT: During the installation of SNAP, do NOT select the option to use SNAP through Python.

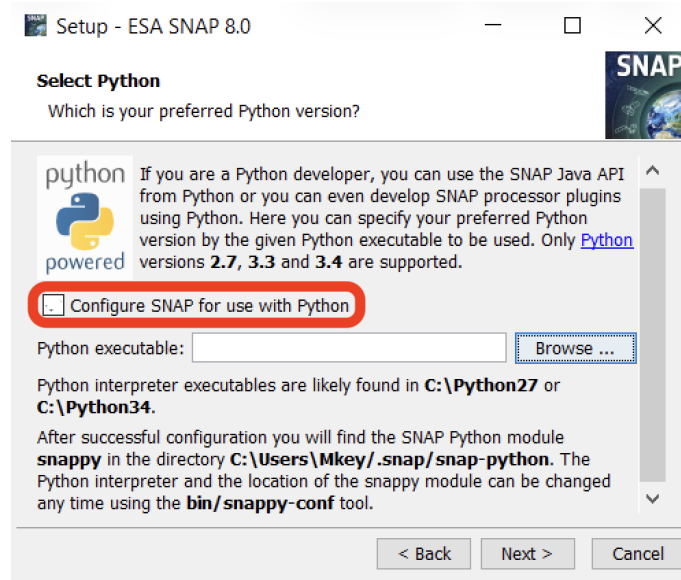


Figure 2.1: Do NOT select *Configure SNAP to work with Python*.

Mandatory plugins

After installation, verify that the following mandatory SNAP plugin module is installed:

- Sentinel-1 Toolbox

Configuration parameters

To ensure stability and performance, you must review the configuration files located in your SNAP installation directory.

1. Increase memory allocation

Edit the file `$HOME/snap/bin/gpt.vmoptions`. Modify the `-Xmx` parameter to match your system's available RAM (e.g., if you have 16GB of RAM, you might set it to 12GB):

```
-Xmx12G
```

2. Optimization settings

Edit the file `$HOME/snap/etc/snap.properties` and ensure the following parameters are set:

```
#snap.home=
#snap.userdir=
snap.jai.tileCacheSize = 1024
snap.jai.defaultTileSize = 512
```

WARNING: Hardware and storage requirements

Processing time-series SAR data requires significant computational

resources. A single Sentinel-1 SLC image is roughly 4 to 8 GB compressed. When unzipped, deburst, coregistered, and processed into interferograms, a standard stack of 40–50 images can easily require 500 GB to 1 TB of free disk space.

Furthermore, depending on your CPU and allocated RAM, the SNAP preprocessing steps (particularly coregistration and interferogram formation) can take anywhere from several hours to several days to complete. Please ensure you have adequate storage space and uninterrupted power before initiating the run.

2.1.2 Python environment

Ensure that a modern Python 3.x environment is installed on your machine. If not, download it and during the installation ensure you check the box to add the Python executable to the environmental variables (PATH) of your OS.

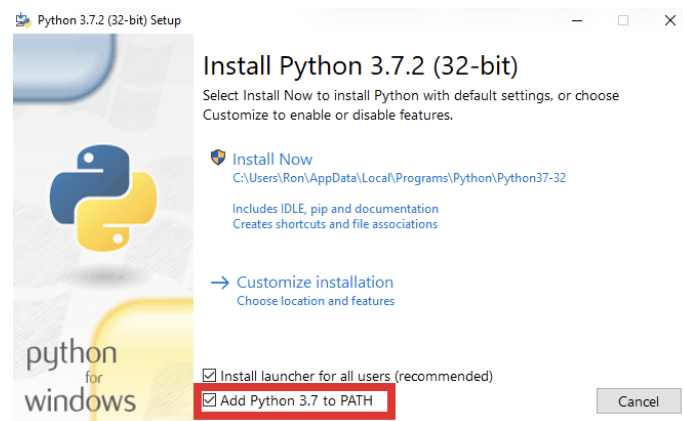


Figure 2.2: How to include Python in the OS PATH on Windows.

The software requires several standard libraries. All required libraries for this tool (such as `os`, `sys`, `subprocess`, `shlex`, `time`, `pathlib`, `shutil`, `json`, `urllib`, and `glob`) are standard Python built-ins and do not require separate installation.

2.1.3 Linux requirements

For users working in a Linux environment, an additional preliminary step is required for the correct functioning of the entire code. It is required to install the software `xterm`. If it is not already done, it can be easily installed by opening the terminal and typing:

```
sudo apt-get install xterm
```

This will be used to open terminal windows during the process when user input is required or to monitor the processing progress.

SUGGESTION: To take advantage of the full computational power of your machine, it is suggested to run this first application in the native OS. Remember that `PHASE_Preprocessing.mlapp` is fully compatible with Windows, macOS, and Linux.

2.2 Required folder structure

The software is provided as a single, unified package requiring a modern Python 3.x environment.

There are no strict requirements for a predefined absolute path or folder structure. The user is free to copy and paste the `PHASE_Preprocessing.mlapp`, the logo images, and the `PHASE_Preprocessing` folder wherever they want on their machine. However, the internal structure of the provided files must not be altered.

Procedure recap:

1. Download the PHASE package.
2. Install all the required software and packages (SNAP 9.x and Python 3.x).
3. Move or extract the downloaded folder into your desired working directory, anywhere on your computer.
4. Execute the `PHASE_Preprocessing.mlapp` MATLAB application.

SUGGESTION: To check the Python version and executable path accessible through your MATLAB software, just type `pyenv` in the MATLAB command window.

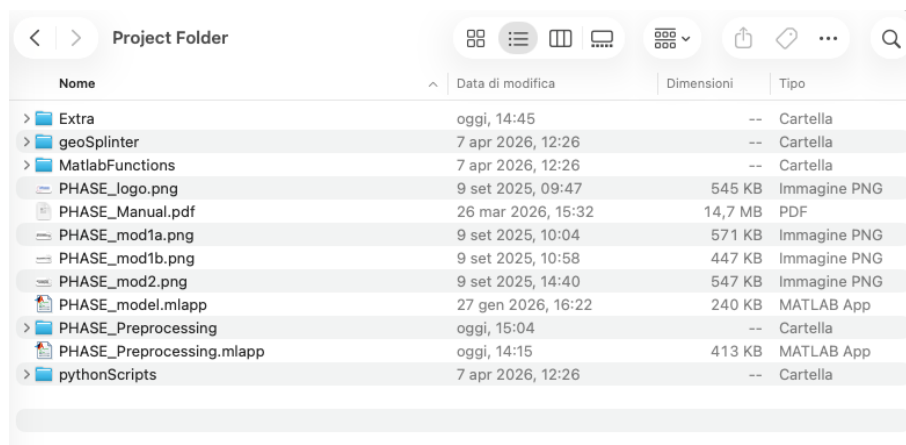


Figure 2.3: Example Project Folder structure.

2.3 Provided snap2stamps files

Inside the `snap2stamps` folder, the core processing architecture is divided into two subfolders: `bin` and `graphs`.

To cleanly distinguish between satellite platforms, all files are strictly prefixed with **SEN** for Sentinel-1 and **CSK** for COSMO-SkyMed.

2.3.1 Python processing scripts (**bin** folder)

The **bin** folder contains the Python 3 engines that drive the automated processing chain. These scripts dynamically generate and execute SNAP processing graphs based on the user's GUI inputs.

SEN_master_selector.py & **CSK_master_selector.py**:

Intelligent Master Selection: Evaluates the temporal baseline of the downloaded stack to find the mathematical median. It then identifies late-summer candidates (August/September) and automatically queries the Open-Meteo historical weather API to select the optimal master image characterized by zero or minimal precipitation over the selected AOI.

SEN_slaves_prep.py & **CSK_slaves_prep.py**:

Initializes and prepares the raw slave image stack for processing.

SEN_splitting_master.py & **CSK_subset_master.py**:

Applies precise orbit files and spatially subsets (or splits) the chosen master image to the exact bounding box defined interactively by the user in the GUI.

SEN_splitting_slaves.py & **CSK_subset_slaves.py**:

Iterates through the slave stack, applying precise orbits and subsetting them to match the master's geometry.

SEN_coreg_ifg_topsar.py & **CSK_coreg_ifg.py**:

Performs the coregistration of the slave images against the master and computes the interferograms. For Sentinel-1, this automatically handles the complexities of TOPSAR debursting and Enhanced Spectral Diversity (ESD) when multiple bursts are detected.

SEN_stamps_export.py & **CSK_stamps_export.py**:

Exports the coregistered stack and interferograms into a binary format strictly compatible with StaMPS PSI ingestion.

SEN_average_intensity.py & **CSK_average_intensity.py**:

Generates the pixel-by-pixel average amplitude image over the chosen AOI.

SEN_terrain_correction.py & **CSK_terrain_correction.py**:

Performs Range-Doppler terrain correction to generate geocoded Local Incidence Angle (LIA) and Coherence (COH) maps for downstream analysis.

2.3.2 Static XML graphs (**graphs** folder)

While most SNAP XML processing graphs are now constructed dynamically in memory by the Python scripts (such as those for ESD multi-burst coregistration and average intensity generation), a few static template graphs remain in the **graphs** folder:

- [SEN_master_split_applyorbit.xml](#) / [SEN_slave_split_applyorbit.xml](#)
- [SEN_export.xml](#) / [SEN_slave_terrain_correction.xml](#)
- [CSK_AOI_subset.xml](#) / [CSK_coreg_ifg_computation.xml](#)
- [CSK_export.xml](#) / [CSK_slave_terrain_correction.xml](#)

Information about External DEM creation and integration is provided in Appendix A.

2.4 Sentinel-1 images

2.4.1 Images download

This first sub-module of PHASE allows the download of Sentinel-1 images from the Alaska Vertex repository (<https://search.asf.alaska.edu>). Here the user can select the AOI by using the available drawing tool. After that, the search must be performed by applying some filters to select the proper input products:

1. Time span: Select the initial and final date of the time span of interest.
2. File type: L1 Single Look Complex (SLC).
3. Beam mode: IW.
4. Polarization: *Leave empty*.
5. Direction: *Choose the orbit depending on your requirements*.
6. Subtype: *Choose the satellite depending on your requirements. For all leave empty*.

Next, inspect the obtained products over the chosen AOI. Among all, click on the one in which the AOI is farther from the borders. Copy the path and frame numbers. Re-open the filters of the search and set them up to further filter your research:

1. Path Start: *Your path number*.
2. Path End: *Your path number*.
3. Frame Start: *Your frame number. If more than one, put the lowest*.
4. Frame End: *Your frame number. If more than one, put the highest*.

Search Filters

Start Date

6/2/2020

MM/DD/YYYY

End Date

3/5/2025

MM/DD/YYYY

Seasonal Search

Additional Filters

L1 Single Look Compl...

IW

Polarization

1/14 file types selected

1/9 beam modes selected

0/8 polarizations selected

Ascending

Subtype

Group ID

1/2 flight directions selected

0/3 subtypes selected

Path and Frame Filters

4

4

25

25

Clear Path / Frame

250 of 142 Files

Cancel

SEARCH

Figure 2.4: Example of applied filters.

Next, you should obtain a list of SAR products all from the same orbit and path for the whole time span. To download all you need to add them to the download queue by clicking on the basket button.

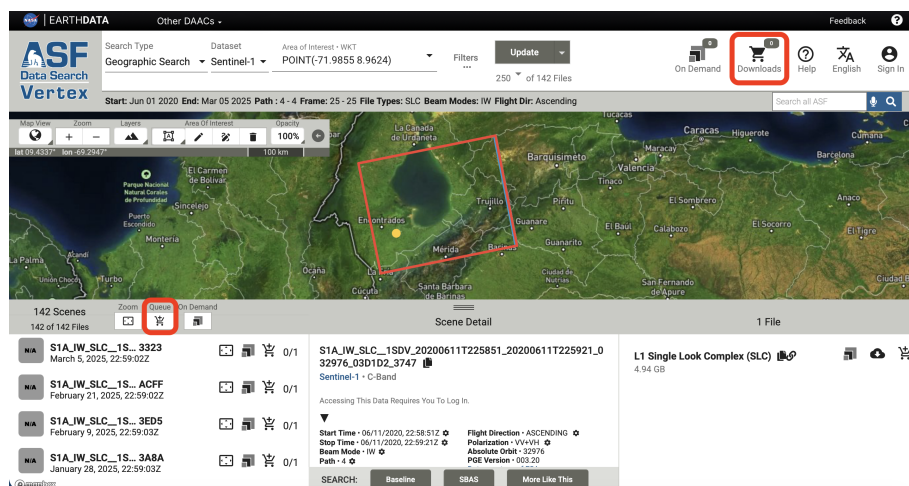


Figure 2.5: Example of obtained product after the filters application.

Finally, click on the basket icon on the top right and on the opened panel click on Data Download and then select Download Python Script (.py). You must have an active account to download the images.

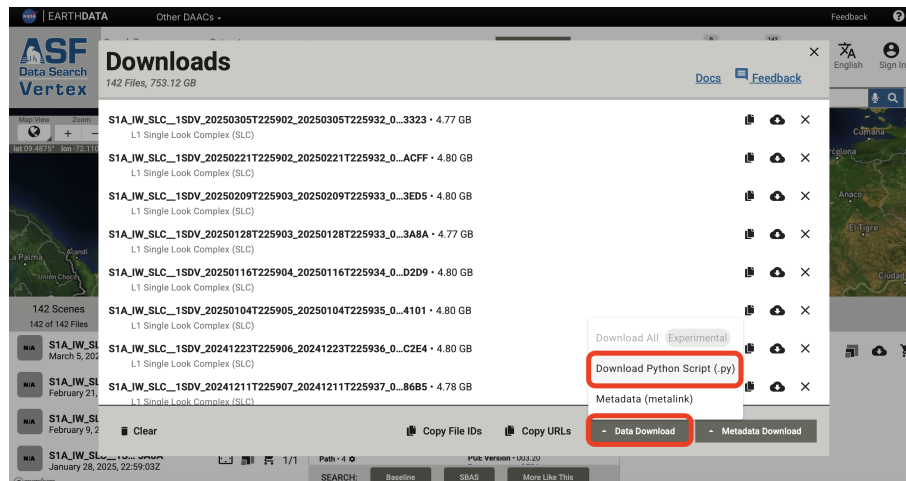


Figure 2.6: Example of Python script download.

Once downloaded, place it in the PHASE_Preprocessing folder, without changing its name.

WARNING: If you want to download all the images “manually” through this Python script follow these steps. Create a `slaves` folder inside the PHASE_Preprocessing folder. Then, open a terminal in the `slaves` folder and write:

```
python download_all_.....py
```

(where `download_all_.....py` is the filename of the Python script you downloaded). Then press enter and the download will start for all your images. At the first use it will ask you the credentials of the EarthData profile (the account you used in the Alaska website).

2.4.2 Area of Interest (AOI) Definition

In previous versions of the software, users were required to manually open reference images in SNAP to identify specific sub-swaths (IW1, IW2, IW3) and burst numbers overlapping their study area. **Starting from PHASE v4.0, this process is entirely automated.**

Once the images are downloaded into the `slaves` folder, the user simply needs to open the `PHASE_Preprocessing.mlapp` interface and navigate to the **AOI** tab.

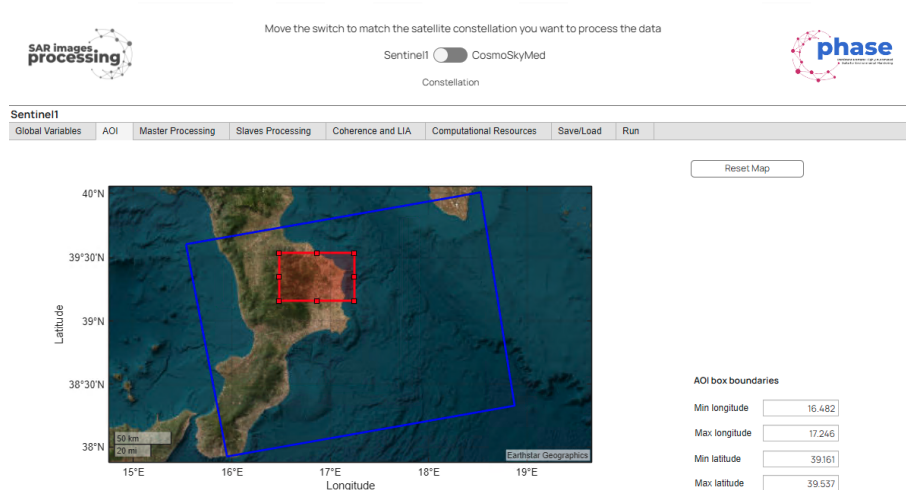


Figure 2.7: Interactive AOI selection using the PHASE graphical map interface.

The software will automatically parse the metadata of the downloaded products and draw a blue polygon on the geographic map representing the full Sentinel-1 footprint. The user can interactively pan, zoom, and drag the red bounding box to define their exact Area of Interest. The coordinates are updated in real-time, and the backend Python scripts will automatically handle the precise spatial sub-setting, debursting, and merging required during processing.

2.4.3 Master image selection

The selection of an optimal master image is critical for minimizing temporal and geometric decorrelation across the interferometric stack. To remove the guesswork, PHASE v4.0 introduces an automated, meteorologically-aware master selection algorithm.

By leaving the **Auto-select master** checkbox enabled in the GUI, the software will:

1. Analyze the temporal distribution of the downloaded stack to find the mathematical median (ensuring temporal centrality).
2. Filter the candidates to prioritize late-summer acquisitions (August and September) to guarantee minimal soil moisture content.
3. Query the **Open-Meteo Historical API** for the selected AOI to evaluate the daily precipitation for the top candidates.
4. Automatically select and stage the driest day (ideally 0.0 mm of rain) closest to the temporal center of the stack.

If the user prefers to force a specific master image (e.g., to align a new processing run with a historical dataset), they can simply uncheck the **Auto-select master** box and manually define the target date using the **Master date** calendar widget.

2.5 COSMO-SkyMed images

2.5.1 Data access proposals

Alongside Sentinel-1 products, PHASE also supports COSMO-SkyMed (CSK/CSG) data. Unlike Sentinel-1, these data are not openly public. Commercial use requires an agreement with e-GEOS, while institutional and research access is granted by the Italian Space Agency (ASI) through specific Open Calls.

To access data for research purposes, researchers must submit a project proposal to ASI:

1. Register an account and create a “Project Card” via the COSMO-SkyMed User Management portal (<https://portal.cosmo-skymed.it/UMUsers/main.html>).
2. Once approved, you can search the catalog and request your authorized data through the CDMFE Portal (<https://portal.cosmo-skymed.it/CDMFE/home>).

2.5.2 Downloading products

Depending on your project’s geographic scope, your download portal and strategy will vary:

- **Italian territory (MapItaly):** If your research covers Italy, you will likely utilize the MapItaly portal (<https://cms.mapitaly.asi.it>). It is strongly suggested to download one of the pre-assembled, automatically created stacks. This ensures that all images share the exact same footprint, guaranteeing successful coregistration and interferometry. Additional information about the MapItaly portal can be found here: <https://www.asi.it/scienze-della-terra/cosmo-skymed/cosmo-skymed-portale-mapitaly/>.

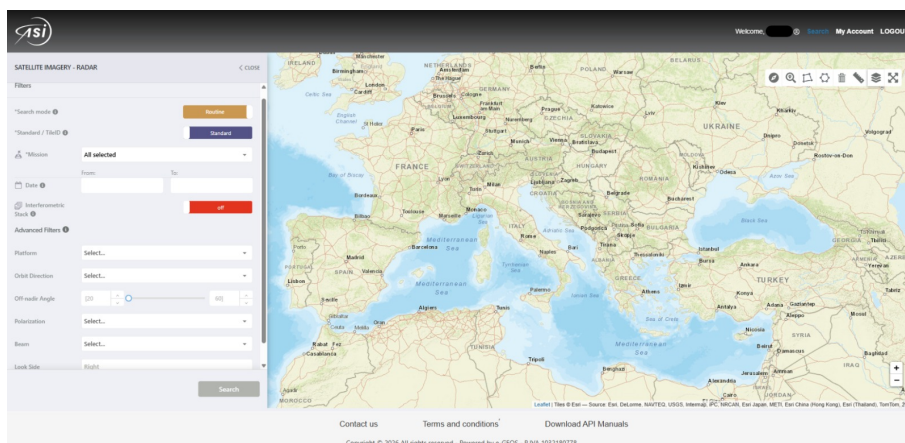


Figure 2.8: Interface of the MapItaly web portal.

- **Global/tasked data:** For regions outside Italy or specifically tasked acquisitions, your approved data will be downloaded directly through the standard CDMFE portal.

2.5.3 Generational compatibility: CSK vs. CSG

The COSMO-SkyMed constellation consists of two satellite generations. Both utilize a stripmap acquisition mode but have distinct file naming conventions and metadata structures:

- **First generation (CSK):** Denoted by filenames starting with CSK (HIMAGE mode).
- **Second generation (CSG):** Denoted by filenames starting with CSG (Stripmap mode).

Starting from PHASE v4.0, the software automatically detects the generation of the input products (CSK or CSG) and dynamically adapts the processing scripts on the fly. Users no longer need to manually swap out Python scripts before running the application.

WARNING: While PHASE seamlessly handles both generations automatically, you **cannot** mix CSK and CSG images within a single processing run due to inherent differences in their orbital geometries and sensor characteristics. Your entire input stack must belong exclusively to one generation.

2.5.4 Area of Interest (AOI) Definition and Subsetting

Unlike Sentinel-1's IW mode—which requires complex splitting to isolate specific swaths and bursts—COSMO-SkyMed acquires data as continuous strips. Because of this, burst and swath parameters are not applicable.

Just like with Sentinel-1, **AOI definition is now entirely interactive through the GUI map**. Once your downloaded .h5 images are placed in the `slaves` folder, open the `PHASE_Preprocessing.mlapp` interface and toggle the constellation switch to COSMO-SkyMed. The software will instantly read the HDF5 root attributes to extract the geodetic coordinates and draw the blue footprint of the images on the geographic map. You can then interactively drag the red bounding box to define your exact spatial subset. The backend processing scripts will automatically crop all images in the stack to match this precise bounding box.

2.5.5 Master image selection

The automated, meteorologically-aware master selection tool introduced for Sentinel-1 is fully supported for COSMO-SkyMed stacks.

If the **Auto-select master** option is enabled in the GUI, PHASE will automatically:

1. Calculate the mathematical median of the time series.
2. Prioritize late-summer acquisitions (August/September) to minimize soil moisture.
3. Query the Open-Meteo historical weather API to select the driest possible day (ideally 0.0 mm of precipitation) closest to the stack's temporal center.

NOTE: If you downloaded a pre-assembled stack from the MapItaly portal, the system may already suggest or mandate a specific master date. In this case, simply uncheck the **Auto-select master** box and manually input the MapItaly-prescribed date into the GUI calendar widget.

2.6 Description of the MATLAB application

The MATLAB application is aimed at providing a simple GUI through which set all the required variables for the processing, as well as checking the progress of the computation. It was tested on MATLAB 2025a running on Windows, macOS and Linux environments. The operations performed by the code are summarized in the following list.

1. **PHASE variables definition:** the user is required to define all the variable parameters needed for both processing and path description. The first step is to select the SAR constellation of the dataset by moving the toggle switch at the top part of the GUI. Several panels are then shown based on the made choice.

- a. **Global Variables:** contains information regarding the Python environment, the download of the SAR images, the time span of the analysis, the type of orbit, and the first and last month.

WARNING: The images download flag works only with the Python script previously downloaded from the alaska.edu portal. Ensure you are using the most recent script version compatible with Python 3.

- b. **Area of Interest (AOI):** provides a geographical map interface where the user can interactively draw and adjust a bounding box over the automatically displayed Sentinel-1 or COSMO-SkyMed footprints.
- c. **Master Processing:** contains all the parameters required for the master configuration file and its date selection. Manual entry of coordinates, swaths, and bursts has been completely removed, as the software now derives them automatically from the interactive AOI map.
- d. **Slaves Processing:** contains the parameters required for the slaves configuration file. There is the possibility to use SNAP built-in DEMs or an external one. The slaves removal flag is intended to declare whether to delete the slave images at the end of the preprocessing to save disk space. Manual swath/burst definition is no longer required.

- e. **Coherence and LIA:** contains a checkbox to toggle this step on or off, and requires the EPSG code for the images projection. Here you can find the code for each region: <https://epsg.io>.
- f. **Computational Resources:** contains the global parameters and path for the correct implementation of the InSAR preprocessing steps.

SUGGESTION: The path to the SNAP gpt executable varies depending on your OS. For Windows, it usually is: `C:\Program Files\snap\bin\gpt.exe`. For macOS, it usually is: `/Applications/snap/bin/gpt`. For Linux, it usually is: `/usr/local/snap/bin/gpt`.

- g. **Save/Load:** allows the user to save the configured variables and/or load them from a previous saving.
- h. **Run:** contains a button to start the execution of the preprocessing steps.

Note that ALL the variables must be configured as required for input by the `PHASE_Preprocessing.mlapp` application. Read the comments for each variable in the panels to better understand their meaning.

2. **Image collection and download:** the necessary SAR images are gathered into the slaves folder. As detailed in the previous sections, Sentinel-1 images can be downloaded automatically by the application using the Python script generated from the ASF Alaska portal. During the first execution, an external command window will open requesting your EarthData credentials. If you have already downloaded the images manually—which is the mandatory procedure for all COSMO-SkyMed data—simply untick the download checkbox in the GUI and ensure all your .zip (Sentinel-1) or .h5 (COSMO-SkyMed) files are placed directly into the slaves folder.
3. **Master image processing:** the selected master image is automatically moved into the master folder. The required folder and subfolder structure for processing the master image is then created. The processing consists of image spatial subsetting and precise orbit correction. The governing parameters depend on the satellite constellation: for Sentinel-1, this involves the automatic identification of the necessary swaths and bursts followed by spatial subsetting; for COSMO-SkyMed, it involves direct spatial subsetting based on the AOI bounding box. The execution phase is composed of two subsequent steps:
 - creation of a .conf file with all the parameters in the correct syntax;
 - execution of the steps through a batch or bash file (for Windows or Linux, respectively) in the system command prompt.
4. **Slave images processing:** this step consists of the standard snap2stamps preprocessing, plus the average scene intensity, and the coherence and local incidence angle estimation. First, the configuration file of the project is

automatically created with the defined information and paths. Then the six steps are executed:

- slaves preparation (step 1);
- slaves splitting and orbit correction (step 2);
- coregistration and interferogram formation (step 3);
- StaMPS export (step 4);
- average scene intensity (step 5);
- LIA and coherence images computation (step 6).

Similarly to the master processing, everything is executed via a batch (or bash) file and an external command window is opened to monitor the processing progress, as it may take a long time. In case you want to reprocess just a single step, you can do so by modifying the initial and final step variables in the Slaves Processing panel. Finally, the folder structure for the StaMPS processing is created. For convenience, the StaMPS processing folder is named with the type of orbit and the initial and final month of the analysis. The `PHASE_StaMPS.mlapp` application does not have to be moved from the `PHASE_Preprocessing` folder, as the software will automatically relocate it to the correct position.

REQUIREMENT: Always select all configurable parameters by clicking on the corresponding drop-down menu; do it also if the interface is already showing your desired value.

WARNING: If during the master processing something goes wrong and during the second execution it is not automatically solved, please **MANUALLY MOVE THE MASTER IMAGE FILE (.zip or .h5)** back to the slaves folder and delete the master folder.

The SNAP operators used in the main processing workflow are reported in the following. The Read and Write operators are sometimes omitted for the sake of brevity.

SENTINEL-1 SNAP OPERATORS

TOPSAR Split → Apply Orbit File
Back-Geocoding → (Enhanced-Spectral-Diversity) → Interferogram
TOPSAR-Deburst → TopoPhaseRemoval → Subset
StampsExport → Terrain Correction

COSMO-SKYMED SNAP OPERATORS

Subset → Create-Stack
Cross-Correlation → Warp
Interferogram → TopoPhaseRemoval → StampsExport → Terrain Correction

Regarding, instead, the developed steps about average intensity, local incidence angle and coherence, the following operators are used.

AVERAGE INTENSITY SNAP OPERATORS

Bands Extractor → Band Merge → Band Maths → Subset → Terrain Correction → Write GEOTIFF

LOCAL INCIDENCE ANGLE SNAP OPERATORS

From Interferogram file: Terrain Correction → Write GEOTIFF

COHERENCE SNAP OPERATORS

From Interferogram file: Terrain Correction → Write GEOTIFF

2.7 Troubleshooting

2.7.1 SNAP external server failures (orbits and DEMs)

During processing, SNAP attempts to automatically download Precise Orbit files and the default DEM from external agency servers. If these servers are undergoing maintenance or your institutional firewall blocks the connection, the processing will fail (often during Step 2 or Step 3).

Solution: Check the official ESA Copernicus or SNAP forums to see if the orbit servers are temporarily down. If the problem persists, you may need to manually download the orbit files or switch to an External DEM (see Appendix A) to bypass the automatic download process.

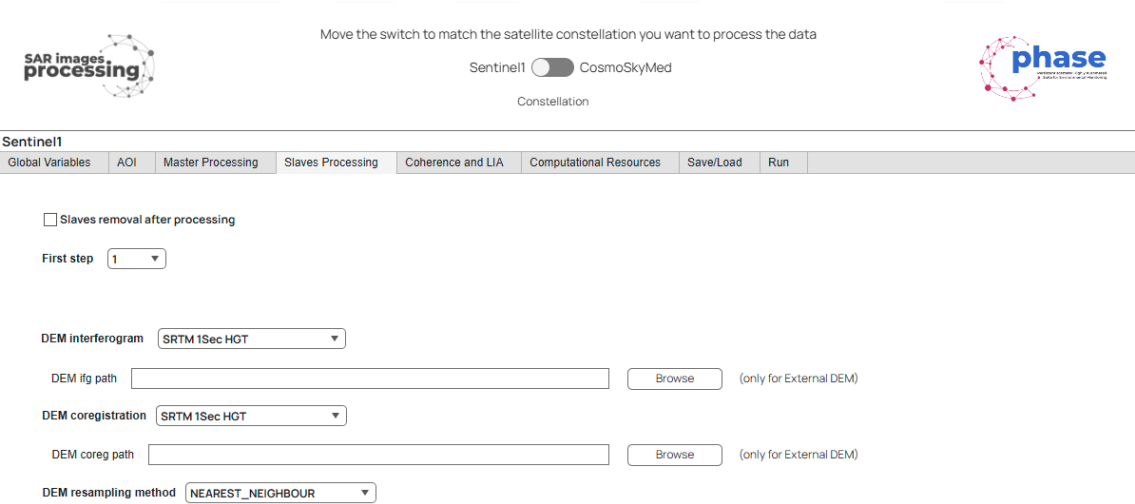
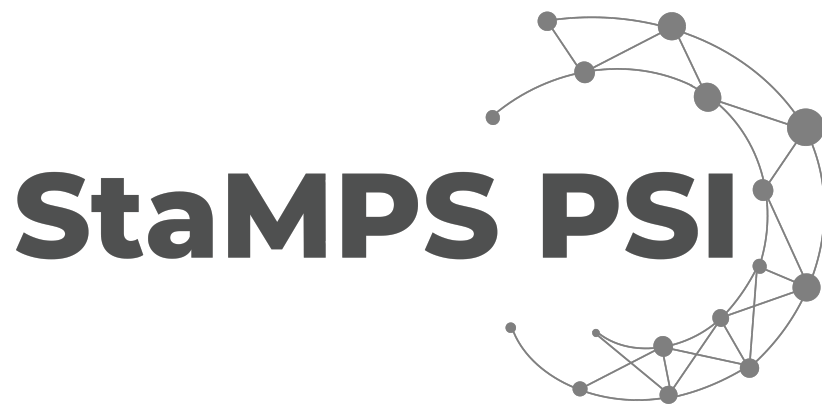


Figure 2.9: PHASE_Preprocessing.mlab application interface.

StaMPS Persistent Scatterer Interferometry



3.1 Preliminary steps

The `PHASE_StaMPS.mlapp` MATLAB application was developed to improve the StaMPS PSI processing. It is designed to automatically perform both the preparation for the StaMPS execution and the execution itself.

3.1.1 Requirements

- **Operating System:** this processing step must be performed in a **Linux environment** due to StaMPS compatibility requirements (Windows is not supported).

- **Environment:** the procedure was tested on Ubuntu 20.04, MATLAB 2025a, and StaMPS 4.1beta (incorporating TRAIN for atmospheric error correction).

3.1.2 StaMPS installation guide

The installation of StaMPS and its dependencies is mandatory. The following steps summarize the standard installation procedure described in the StaMPS V4.1b manual.

1. Download and compilation

Download the source code (v4.1b or later) from the official GitHub repository. Open your terminal and execute the following commands to compile the source code:

```
# Extract the tarball (if downloaded as .tar)
tar -xvf StaMPS_v4.1b.tar
```

```
# Navigate to the source directory
cd StaMPS_v4.1b/src
```

```
# Compile and install
make
```

2. Required dependencies

Snaphu: The 3-D unwrapping algorithms used in StaMPS require **Snaphu**. It must be installed separately and can be downloaded from Stanford Radar Software.

3. Optional packages

- **TRAIN:** The Toolbox for Reducing Atmospheric InSAR Noise (TRAIN) is highly recommended for correcting tropospheric delays. Download it from GitHub.
- **Triangle:** Required only if you intend to run Step 8 of the standard StaMPS processing (Delaunay triangulation). It is not required for the standard Steps 1–7.

4. Configuration

StaMPS requires specific environment variables to be set. Locate the configuration file named `StaMPS_CONFIG.bash` (or `.tcsh` if using C-shell) in the installation folder.

Open the file and edit the directory paths to match your system setup. The figures below illustrate the folder structure and configuration used by the authors.

IMPORTANT: You must **edit** the `StaMPS_CONFIG.bash` file to set the correct paths. However, you do **not** need to source it manually (e.g., in your `.bashrc`); PHASE automatically sources this file before every execution.

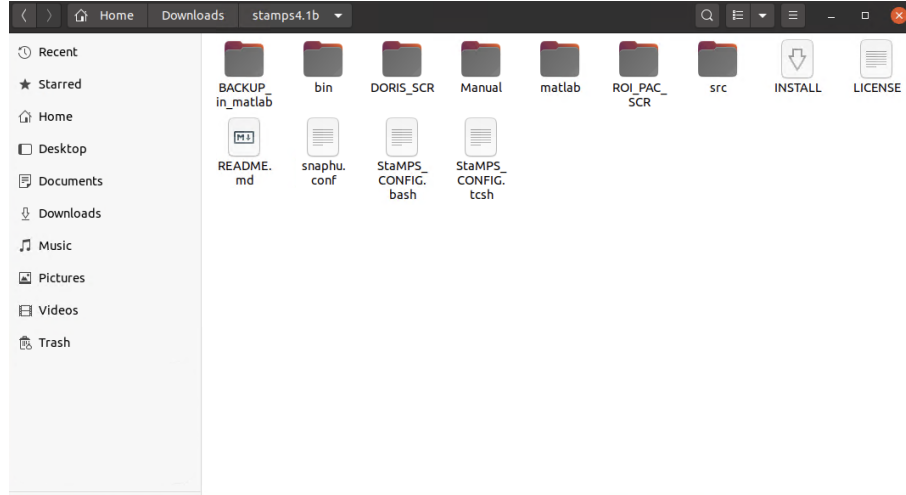


Figure 3.1: Example StaMPS folder setup.

NOTE: While not mandatory, the authors have StaMPS and its add-ons installed in the `Downloads` folder of the Linux environment (see example above). Ensure there are no duplicate filenames in the installation directory.

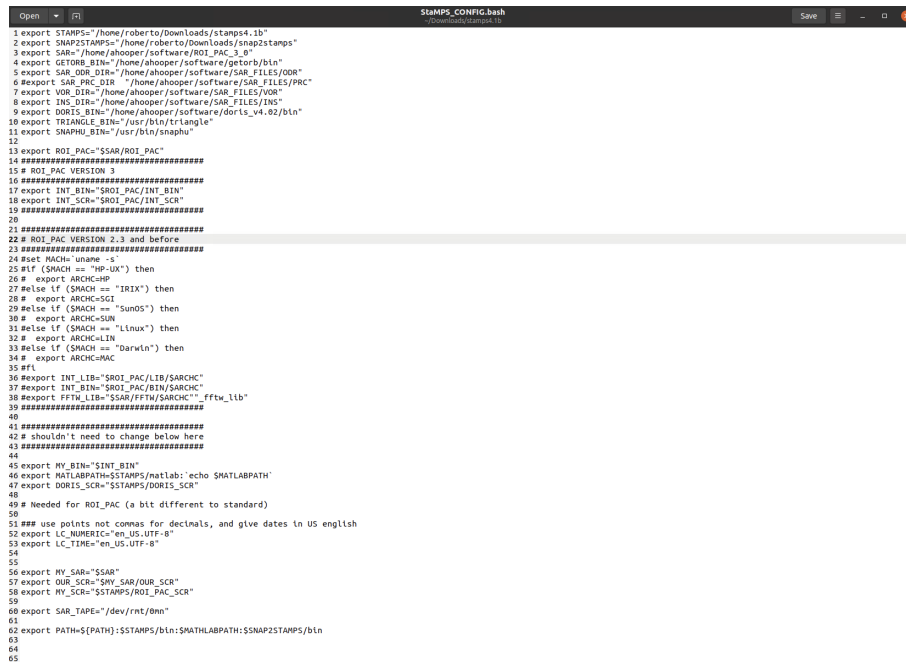


Figure 3.2: Example `StaMPS_CONFIG.bash` file configuration.

3.2 Required folder structure

The provided MATLAB application (PHASE_StaMPS.mlapp) must remain in the PHASE_Pre-processing folder during the execution of the preprocessing sub-module. The software automatically handles file relocation during that execution.

CRITICAL REQUIREMENT: During the first pre-processing run, the software automatically creates a new project folder and moves the .mlapp file into it.

Target Folder Name Format: ORB_START_END
(e.g., ASC_Aug22_Sep26)

- **ORB:** Orbit type (ASC or DSC).
- **START:** Month/Year of first observation (e.g., Aug22).
- **END:** Month/Year of last observation (e.g., Sep26).

Verification: Check that PHASE_StaMPS.mlapp has moved into this new folder inside your project directory. **If the file was not moved automatically, you must move it manually before proceeding.**

3.3 Description of the MATLAB application

The code automates the StaMPS PS analysis in three subsequent steps:

1. **Preparation:** preparing required data for StaMPS processing.
2. **Definition:** setting all StaMPS software parameters.
3. **Execution:** running StaMPS and exporting displacement time series to Excel.

3.3.1 Application panels

The GUI contains several panels for configuration:

- a) **Preparation:** this panel contains the generic paths to the project folder and the one where StaMPS is installed.

REQUIREMENT: Do not select the StaMPS or PHASE_Preprocessing folders directly. Instead, select the **directory that contains them**.
Example: If installed in your Downloads folder, set the path to Downloads, not Downloads/StaMPS.

In addition to that, this panel defines the amplitude dispersion threshold.

Automated Parameter Detection: Upon initialization, the software automatically scans the `INSAR_...` directory to extract the Master Date. If successful, the Master Date field is automatically populated and locked to prevent user error. Manual input is only required if the folder structure is invalid.

Execution Logic: Check the box to run the StaMPS preparation step.

- **Initial Run:** mandatory.
 - **Reprocessing:** optional (can be unchecked).
- b) **Global Variables:** this panel contains all the parameters required for the correct definition of the StaMPS input and output files. Similar to the master date, the software parses the `.diff.par` metadata files to automatically detect and lock the Time Span, Heading, Radar Wavelength (λ), and Center Time. If the auto-detection fails (e.g., missing metadata files), the fields will remain unlocked for manual input.
- c) **StaMPS 1 – StaMPS 5:** these panels contain all the parameters required for the correct definition of each step of the StaMPS processing. The TRAIN box is intended to declare if to use the TRAIN software or not. Two TRAIN models are currently implemented in the code: linear and GACOS.

Below is a detailed description of the key parameters available for configuration in these panels, derived from the official StaMPS manual:

Step 2: Estimate Phase Noise

- `max_topo_err`: maximum uncorrelated DEM error (in meters). Pixels with error greater than this are excluded.
- `filter_grid_size`: pixel size of the grid (in meters) used for re-sampling before filtering to determine spatially-correlated phase.
- `clap_win`: window size for the Combined Low-pass and Adaptive Phase (CLAP) filter.

Step 3: PS Selection

- `select_method`: method for pixel selection. Options are DENSITY or PERCENT.
- `density_rand`: maximum acceptable spatial density (per km^2) of selected pixels with random phase. Applies only if DENSITY is selected.
- `percent_rand`: maximum acceptable percentage of selected pixels with random phase. Applies only if PERCENT is selected.

Step 4: PS Weeding

- `weed_standard_dev`: threshold standard deviation. Pixels with a noise standard deviation greater than this value are dropped.
- `weed_time_win`: smoothing window (in days) for estimating phase noise distribution.

Step 5: Phase Correction

- `merge_resample_size`: coarser posting (in meters) to resample to. If set to 0, no resampling is applied.
- `merge_standard_dev`: threshold standard deviation for resampled pixels.

Step 6: Phase Unwrapping

- `unwrap_method`: the unwrapping method to apply.
- `unwrap_grid_size`: resampling grid spacing (in meters). Phase is resampled to this grid if pre-filtering is enabled.
- `unwrap_gold_n_win`: window size for the Goldstein adaptive phase filter.

Step 7: SCLA Estimation

- `scla_deramp`: if set to `y`, a phase ramp is estimated for each interferogram.
- `subtr_tropo`: if set to `y`, removes a tropospheric delay estimate before SCLA computation. See Section 3.3.2 for additional details.

d) **StaMPS AOI**: this panel contains the variables for defining the spatial reference area for phase unwrapping, as well as parameters for data visualization. Setting a proper reference point is critical, as all displacement measurements in your final output will be calculated relative to this location.

- **Reference longitude / latitude**: these fields require the geographic coordinates of the specific location you want to act as your “zero displacement” reference (ideally, you should choose a known stable area, like exposed bedrock or a structurally sound building). If these fields are left empty, the software will default to using the spatial average of the entire processed scene as the zero reference.
- **Reference radius**: rather than relying on a single, potentially noisy Persistent Scatterer to act as your zero point, this parameter defines a circular area (radius) around your defined Reference Lon/Lat. The software will identify all PS points falling within this radius and average their values together to establish a much more robust and stable zero-reference.
- **PS plot size** (`ps_plot`): this is strictly a visual parameter for the MATLAB interface. It controls the marker size of the individual scatterers when the software generates the interactive displacement plots. It does not affect the underlying data or calculations in any way.

e) **Computational Resources**: this panel contains the global parameters and path for the correct implementation of the InSAR preprocessing steps.

f) **Save/Load**: this panel allows the user to save the configured variables and/or load them from a previous saving.

- g) **Run:** this panel contains a button to start the execution of the processing steps.

Note that all the variables must be configured as required as input by the PHASE_StaMPS.mlapp software. Read the comments for each variable to better understand their meaning.

WARNING: The usage of this code is based on the previous installation of StaMPS and all its additional packages as specified in the StaMPS user manual. Please ensure that during the data preparation step, all interferograms have a mean amplitude greater than zero and no warnings arise. Otherwise, you must remove the problematic interferograms before StaMPS processing to avoid computational errors.

WARNING: This code contains a `keyboard` command after the automatic creation of the interactive PS plot where the user has to choose the radius and central point for the data export. It is suggested to set a very large radius and select the center of the figure in case all the PS points are relevant for the export. Unfortunately, there is no way to overcome this manual intervention of the user. In addition to that, when also using the TRAIN toolbox, it may be required to press the `continue` button of the MATLAB editor panel during step 7. Note that this is something StaMPS-related and therefore out of the control of PHASE.

3.3.2 Tropospheric correction (TRAIN integration)

When the `subtr_tropo` parameter is set to `y` and the TRAIN toolbox is successfully installed, PHASE can correct for tropospheric phase delays before the final phase unwrapping. The application currently supports two atmospheric correction models, which can be selected via the GUI:

- **Linear model (`a_linear`):** this method estimates a linear relationship between the interferometric phase and the topography (elevation) for each interferogram. It does not require any external weather data. If selected, the script will automatically run the `aps_linear` TRAIN function after Step 6 and apply the corrections seamlessly without user intervention.
- **GACOS model (`a_gacos`):** this method utilizes high-resolution tropospheric delay maps provided by the Generic Atmospheric Correction Online Service (GACOS). Because GACOS data must be requested externally, this method requires a brief manual intervention during processing.

If `a_gacos` is selected, PHASE will perform the following sequence:

1. **Execution pause:** the processing will run up through StaMPS Step 6 and then automatically pause (triggering a `keyboard` command in MATLAB).

2. **Information file:** PHASE will generate a text file in your project folder named `GACOS_download_info.txt`, containing the exact dates, times, and bounding box coordinates required for your dataset.
3. **Data request:** open your web browser, navigate to `www.gacos.net`, and submit a new request using the precise parameters listed in the text file.

CRITICAL: When filling out the GACOS request form, you must select the **binary** output format. Do not select GeoTIFF, as the TRAIN software cannot process GeoTIFFs.

4. **Download & 20-date limit:** GACOS limits requests to a maximum of 20 dates per request. If your time series contains more than 20 dates, you must submit multiple requests and wait to receive all the corresponding download links via email.
5. **Placement:** Place **ALL** the downloaded compressed folders (`.tar.gz`) directly into the newly created GACOS folder inside your project directory.

NOTE: Do not extract or repackage the archives. The software natively supports multiple `.tar.gz` files and will process them in bulk.

6. **Resume processing:** return to MATLAB and press the `Continue` button in the Editor tab (or type `return` in the Command Window and press Enter). PHASE will automatically extract all archives, sort the `.ztd` and `.ztd.rsc` files into their correct date-specific subdirectories, clean up the temporary extraction folders, compute the corrections, and seamlessly finish the remaining StaMPS steps.

3.3.3 Tuning PS density (how to increase PS points)

If your initial analysis results in a sparse distribution of Persistent Scatterers and you need to increase the final number of PS points, you can relax the selection criteria in two different phases of the processing:

1. **Amplitude dispersion threshold (Preparation panel):** by increasing this threshold (e.g., from the standard 0.40 to 0.42 or 0.45), you allow the algorithm to select more pixels as initial PS Candidates (PSC) before the phase analysis even begins.
2. **Phase noise threshold (Step 3 panel):** depending on your already chosen `select_method`, you can increase either the `percent_rand` or the `density_rand` parameters. Raising these values allows the algorithm to retain a higher percentage or density of pixels that exhibit random phase characteristics.

WARNING: There is a strict trade-off between PS density and data quality. While increasing these parameters will give you a denser grid of points, it inherently lowers the signal-to-noise ratio of your dataset by including lower-quality, noisier scatterers. This can introduce severe errors during the phase unwrapping step (Step 6).

3.4 Outputs

The software generates an Excel (.xlsx) and a CSV (.csv) spreadsheet containing the displacement (in mm) for each PS at each SAR image epoch. Alongside the displacement data, a supplementary file containing the atmospheric modeling used during the processing is also provided.

The output files are structured according to the format shown in the reference table below. Please note that the displacement at the master date is explicitly set to zero for all points, as it serves as the temporal reference for the entire time series.

Table 3.1: Example output format

Point	Lon	Lat	Vel [mm/yr]	Date 1	Date 2	...
1	9.123	45.456	-2.5	0.0	-1.2	...
2	9.124	45.457	-1.8	0.0	-0.9	...

3.5 Troubleshooting

Common errors unrelated to the PHASE software but dependent on external libraries are listed below, together with their solution.

3.5.1 StaMPS step 5 error

During Step 5, you may encounter the following error:

```
STAMPS: #####
STAMPS: ##### Step 5 #####
STAMPS: #####
STAMPS: Directory is PATCH_1

PS_CORRECT_PHASE: Starting
Correcting phase for look angle error...
GETPARM: small_baseline_flag='n'
PS_CORRECT_PHASE: Finished
Error using movefile
Text file busy

Error in stamps (line 462)
    movefile('patch.list','patch.list_old');
```

Solution: Open `stamps.m` and change the command `movefile` to `copyfile` at lines 462 and 463.

3.5.2 GCC and G++ compatibility

StaMPS requires `gcc-7` and `g++`. Newer Ubuntu versions (e.g., 22.04) are not natively compatible.

Solution:

- Use an older Ubuntu version (e.g., 20.04).

- Or, manually install `gcc-7` on the newer OS (follow the procedure from AskUbuntu or similar forums). There might be other packages required to successfully download `gcc-7` and `g++`.

3.5.3 Folder navigation error

If you wish to process SAR images in a different folder after a previous run, you must:

1. Close MATLAB completely.
2. Navigate to the new folder.
3. Open `PHASE_Preprocessing.mlapp` again.

This prevents "current directory" errors.

3.5.4 SNAP compatibility (mt_prep_snap)

There are known incompatibilities between data prepared with SNAP 10.x and StaMPS. Newer versions of SNAP were not investigated yet.

Requirement: Please stick to **SNAP 9.x**.

StaMPS

Preparation Global Variables StaMPS 1 StaMPS 2 StaMPS 3 StaMPS 4 StaMPS 5 StaMPS AOI Save/Load Run

☒ data preparation for StaMPS

StaMPS installation folder

Path Browse

Project folder

Path Browse

Master date 05-Mar-2020 ▼

Amplitude threshold

Value 0.40

StaMPS PSI phase

Figure 3.3: `PHASE_StaMPS.mlapp` application interface.

Module 2

Spatio-temporal modeling of PSI displacements



4.1 Preliminary steps

The `PHASE_model.mlapp` MATLAB application was developed to perform interpolations and modeling of PS time series generated by the first module of PHASE. However, it is highly versatile and designed to work with any external SAR data, provided it is formatted in the identical tabular structure.

The application requires MATLAB and a recognized Python 3 environment, specifically utilizing the `openpyxl` Python module to generate the automated Excel reports.

SUGGESTION: To check the Python version and environment path currently accessed by your MATLAB software, type `pyenv` in the MATLAB command window. You can install the required Python module

by opening your OS terminal or command prompt and running: `pip install openpyxl`.

Some geospatial modeling operations can be highly computationally expensive. If the execution takes too long or exceeds your system's memory, you should try reducing the spatial resolution of the requested output grid.

WARNING: Please remember that all provided InSAR displacements are measured strictly along the satellite Line-of-Sight (LOS), which is a 1D vector. The software models this specific 1D deformation and does not inherently decompose it into true vertical or horizontal 3D vectors.

This module of PHASE requires the following MATLAB packages to be installed:

- Curve Fitting Toolbox.
- Image Processing Toolbox.
- Mapping Toolbox.
- Optimization Toolbox.
- Parallel Computing Toolbox.
- Signal Processing Toolbox.
- Statistics and Machine Learning Toolbox.

4.2 Required folder structure

The provided `PHASE_model.mlapp` application is already distributed in the correct project folder location, meaning the user does not need to move it. It must remain in the main project folder, alongside `PHASE_preprocessing.mlapp`, so that it can automatically locate and access the output directories generated by the first module.

4.3 Description of the MATLAB application

PHASE Module 2 takes the raw, scattered PS time series generated by Module 1 (or external PSI processors) and automatically converts them into continuous models and actionable reports. Each panel of the provided MATLAB application is below described.

4.3.1 Input files

Here you need to set the paths to the PSI output spreadsheet (either `.xlsx` and `.csv`) and to Python.

Additionally, the date of the first observation in the time series must be specified. There is also a checkbox to automatically remove all the previous output folders prior to the software execution.

4.3.2 Area of interest

The second step involves defining the AOI, which delineates the spatial boundaries for computing and modeling displacements. This AOI can be specified either through a shapefile (recommended for its flexibility in accommodating arbitrary shapes) or a simple bounding box defined by minimum and maximum longitude and latitude coordinates. The code automatically detects whether the input coordinates are in a projected system (e.g., UTM) or geographic (lat/lon) and handles any necessary transformations to ensure consistency. For shapefiles, it reads the geometry, checks the coordinate reference system (CRS), and converts between systems if needed. Points outside the AOI are filtered to focus processing only on relevant PS.

4.3.3 Processing options

Depending on your analytical needs, the application allows you to choose between two alternative main processing modes, both of which conclude with automatic report generation:

Alternative 1: Temporal-only modeling (PS-by-PS)

This mode focuses strictly on the time domain without performing any spatial interpolation between points. The software automatically cleans and mathematically decomposes each individual PS time series. This includes: outlier removal, low-degree polynomial trend estimation, Fourier-based periodicity analysis, spline-based modeling of residual local components, and stochastic collocation on the residuals.

Alternative 2: Spatio-temporal modeling (interpolation)

This mode performs a joint spatio-temporal modeling, taking the scattered points data and interpolating them into continuous displacement fields. You can choose between two mathematical approaches (Deterministic splines or Stochastic Least Squares Collocation) across two spatial domains:

- **1D domain:** used when monitoring linear structures defined by a centerline (e.g., roads, railways, narrow arch dams).
- **2D domain:** used when monitoring wide, areal targets (e.g., cities, landslides, earth-filled dams).

Final step: Automatic reporting

Regardless of which modeling path you choose, the module finalizes the

process by generating a comprehensive Excel report (via Python integration) and reproducible, publication-ready maps. It also automatically applies thresholding to highlight specific areas exhibiting anomalous displacement behavior.

The overall flow of these processing options is schematized in the figure below.

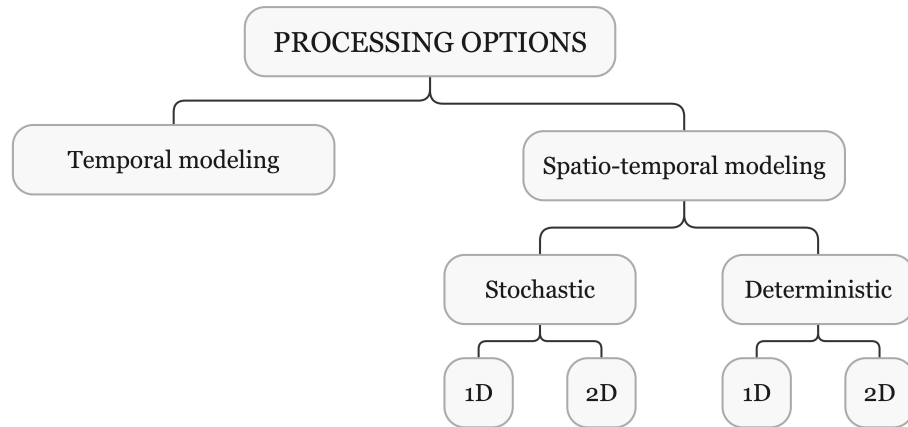


Figure 4.1: Schematic representation of the processing options available in PHASE.

Consequently, the GUI has 4 available processing types:

- **Temporal**: interpolation of all time series independently.
- **Temporal&NNI**: interpolation of all time series independently and nearest neighbor spatial interpolation for visualization.
- **Spatio-temporalDET**: spatial interpolation with deterministic methodology.
- **Spatio-temporalSTC**: spatial interpolation with a joint deterministic and stochastic methodology.

The following subsections explain how to configure each tab in the GUI.

4.3.4 Temporal modeling

When the user selects the temporal-only mode, Module 2 performs a fully automated decomposition of every individual PS time series. The objective is to separate the physical deformation signal from background noise by modeling each scatterer independently, without influence from neighboring points.

The configuration panel

The entire temporal workflow is driven by a dedicated GUI tab that centralizes all user-configurable parameters.

Figure 4.2: Configuration panel for the temporal modeling workflow in PHASE. Users select the main options (noise model, spline knots, regularization, collocation type) from drop-down menus and numerical fields.

Through this interface, users can define the core parameters that govern the underlying mathematical estimators without needing advanced statistical expertise. Key selectable options include:

- **A priori noise variance model:** choose between a coherence-based model, a constant manual value, or automatic estimation from the Fourier spectrum.
- **Spline knot number:** allow the software to automatically estimate the number of knots via statistical methods (variance threshold, MDL, F-test, χ^2 test), or define them manually.
- **Regularization strength:** set the smoothing parameter λ for the spline fit (automatic or user-defined).
- **Collocation strategy:** choose between filtering (estimation strictly at observation points) or prediction (interpolation).
- **Centerline/grid resolution:** define the resolution of the regular spatial domain for the estimation (*available only if NNI is selected*).

The processing workflow

Once initiated, the software processes each PS through five sequential core stages, plus one optional spatial visualization stage.

TEMPORAL MODELING

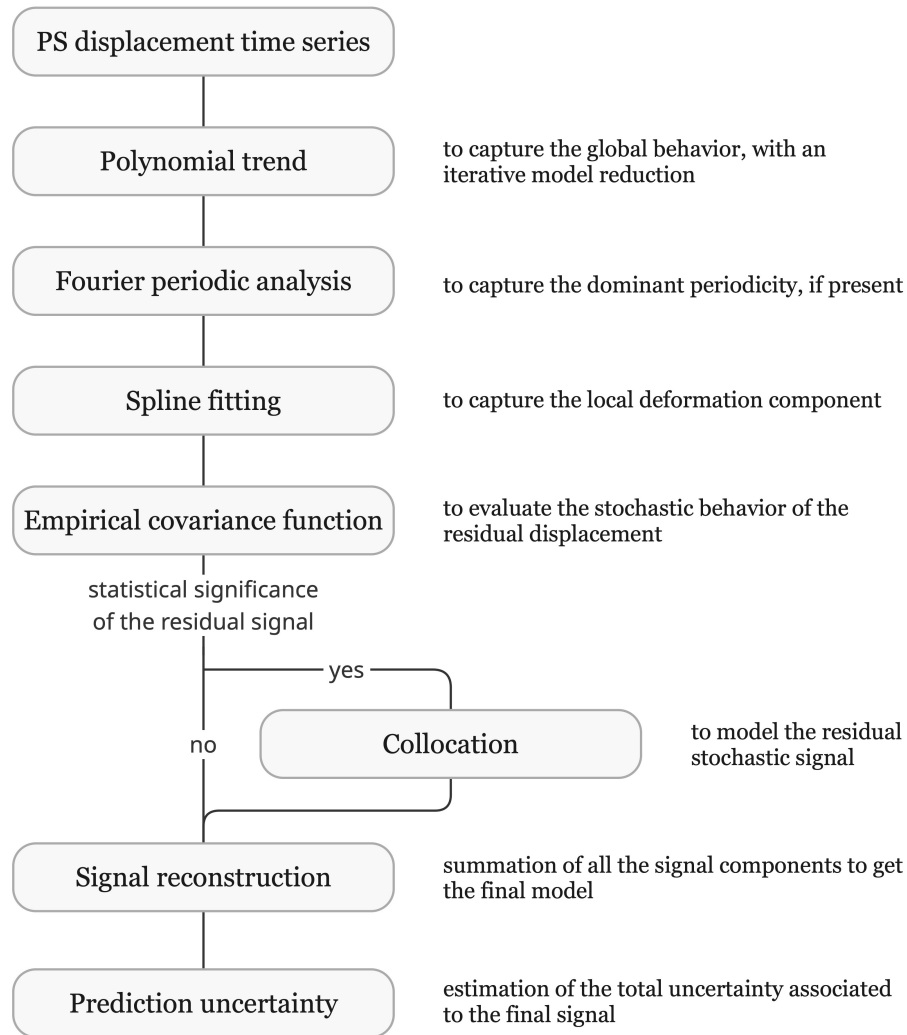


Figure 4.3: Schematic representation of the temporal modeling workflow. Each PS time series is decomposed into deterministic, periodic, local, and stochastic components.

1. **Deterministic trend estimation:** isolates the long-term linear or polynomial deformation component.
2. **Periodic/seasonal analysis:** identifies and isolates cyclic thermal or seasonal oscillations.
3. **Local modeling with splines:** captures non-linear, localized kinematic behaviors.
4. **Empirical covariance estimation:** analyzes the residual stochastic (random) process.

5. **Collocation:** applies filtering or prediction to obtain a smoothed, physically consistent time series.
6. **Optional spatial visualization:** generates a quick spatial overview using Nearest Neighbor Interpolation (NNI).

Final signal reconstruction

Because the model is built sequentially, the final reconstructed displacement estimate $\hat{d}(t)$ for any given PS is simply the sum of the isolated physical components evaluated at the target epochs:

$$\hat{d}(t) = \hat{d}_{\text{trend}}(t) + \hat{d}_{\text{periodic}}(t) + \hat{d}_{\text{spline}}(t) + \hat{d}_{\text{collocation}}(t) \quad (4.1)$$

Understanding uncertainty: Because each component is estimated conditionally based on the previously removed signals, you cannot simply add their individual standard deviations together. Therefore, the total uncertainty of the fully reconstructed time series $\sigma_{\hat{d}}(t)$ is represented solely by the point-wise prediction variance of the final modeled residuals.

Depending on whether the stochastic signal is physically significant enough to trigger collocation, the final reported uncertainty is defined as:

$$\sigma_{\hat{d}}^2(t) \approx \begin{cases} \sigma_{\hat{d}_{\text{spline}}}^2(t) & \text{if stochastic signal is negligible} \\ & \text{(collocation skipped)} \\ \sigma_{\hat{d}_{\text{collocation}}}^2(t) & \text{otherwise} \end{cases} \quad (4.2)$$

4.3.5 Spatio-temporal stochastic

The stochastic approach evaluates the behavior of all PS points simultaneously. This allows the software to rigorously filter out spatially uncorrelated noise (e.g., phase unwrapping errors) and statistically interpolate data across areas lacking radar targets.

The complete procedural sequence for this stochastic spatio-temporal modeling is schematized in Figure 4.4. This is valid for both 1D and 2D geometries.

SPATIO-TEMPORAL STOCHASTIC MODELING

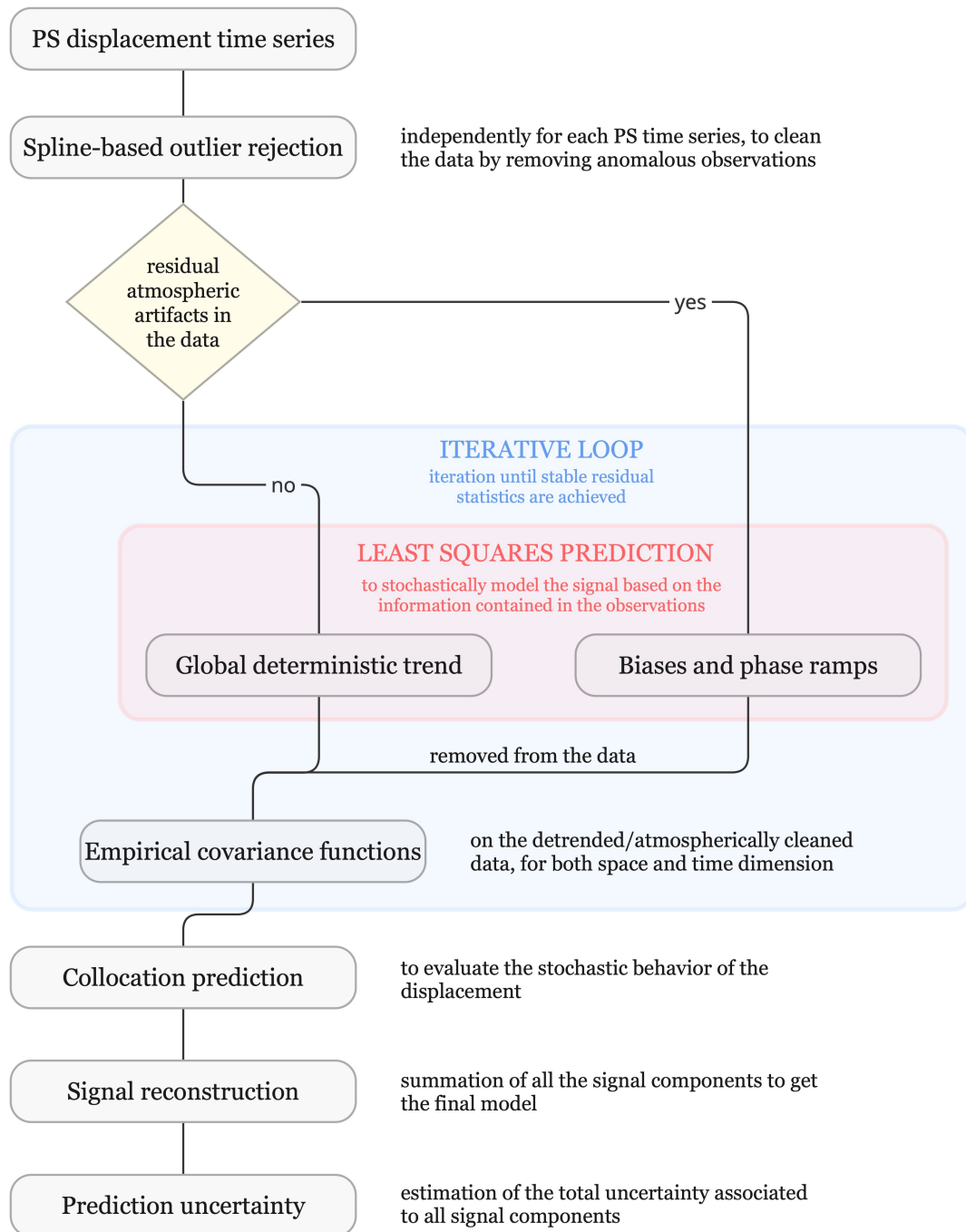


Figure 4.4: Schematic representation of the stochastic spatio-temporal modeling workflow, progressing from raw PS observations through outlier rejection, empirical covariance fitting, and LSC.

To facilitate the interpretation of this massive, multi-dimensional dataset, the software automatically generates dynamic and static visualizations as the final

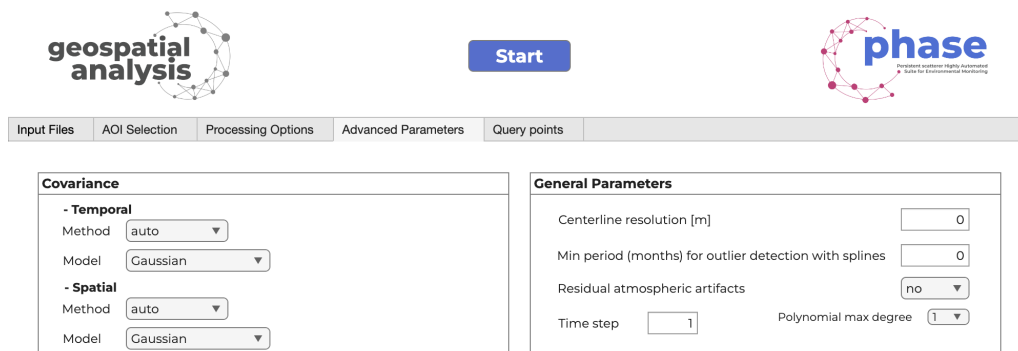
step of the processing workflow:

Dynamic time-lapse (GIF): a continuous animation is produced displaying the reconstructed displacement field superimposed on a satellite basemap. This allows users to visually track the progression of deformation phenomena throughout the entire monitoring period.

Static spatio-temporal maps: *only for 1D spatial geometry* to fit both the spatial and temporal dimensions into a single plot, the software generates 2D raster maps where the X-axis represents the curvilinear centerline coordinate (ℓ), the Y-axis represents time (t), and the color scale represents the continuous displacement magnitude. An identical accompanying map is generated for the estimation uncertainty ($\sigma_{\hat{d}}$), providing a quantitative assessment of the model reliability across space and time.

1D spatial domain

The entire workflow is configured through a dedicated graphical interface tab (Figure 4.5), providing direct control over the underlying statistical estimators without requiring manual code modifications.



The configuration panel for the 1D stochastic modeling workflow in PHASE. It features a 'geospatial analysis' logo, a 'Start' button, and a 'phase' logo. Below these are tabs for 'Input Files', 'AOI Selection', 'Processing Options', 'Advanced Parameters', and 'Query points'. The 'Advanced Parameters' tab is active, showing two main sections: 'Covariance' and 'General Parameters'. The 'Covariance' section has sub-sections for '- Temporal' and '- Spatial', each with 'Method' (auto) and 'Model' (Gaussian) dropdowns. The 'General Parameters' section includes 'Centerline resolution [m]' (0), 'Min period (months) for outlier detection with splines' (0), 'Residual atmospheric artifacts' (no), 'Time step' (1), and 'Polynomial max degree' (1).

Figure 4.5: Configuration panel for the 1D stochastic modeling workflow in PHASE. Users define the spatial/temporal prediction grids, outlier rejection thresholds, trend models, and covariance fitting parameters.

Through this interface, you govern the specific estimators used in the Least Squares Collocation framework. Key configurable parameters include:

- **Centerline and time resolution:** defines the uniform spatial ($\Delta\ell^{\text{out}}$) and temporal (Δt^{out}) step sizes for the target prediction grid. This dictates the resolution of your final continuous displacement model.

- **Minimum period for splines fitting:** establishes the maximum flexibility of the spline curves used during the preliminary temporal outlier rejection step, preventing the algorithm from overfitting short-term noise.
- **Approach for LSC (Signal vs. Noise separation):** dictates how large-scale atmospheric delays and orbital errors are handled. You can choose to absorb them into the noise model, or assume clean data and fit a low-degree deterministic polynomial prior to collocation.
- **Atmospheric residual errors:** if present, specifies whether these errors are treated as purely time-variable biases (suitable for small AOI) or as spatio-temporal trends (required for large AOI).
- **Global polynomial:** defines the maximum permissible degree for the global deterministic trend fitted across the entire infrastructure axis (if assuming clean data).
- **Steps to average the empirical covariance:** controls the lag-binning resolution used to compute the empirical covariance point clouds. This can be determined automatically based on data density or set manually.
- **Covariance model:** selects the continuous mathematical function used to fit the empirical covariance data. This generates the positive-definite matrix strictly required for the final stochastic prediction.

To construct a continuous spatio-temporal deformation model along a linear infrastructure, the irregularly scattered PS observations must first be orthogonally projected onto the centerline of the element. Their spatial position is subsequently described purely by a 1D curvilinear coordinate (ℓ), originating at the beginning of the selected segment.

WARNING: If portions of the infrastructure lack PS points (e.g., highway tunnels, which are undetectable by SAR), these sections are treated mathematically as data gaps. Because this model generates a continuous field, displacement values within these gaps are the result of pure statistical extrapolation. Always interpret extrapolated values in large data gaps with caution.

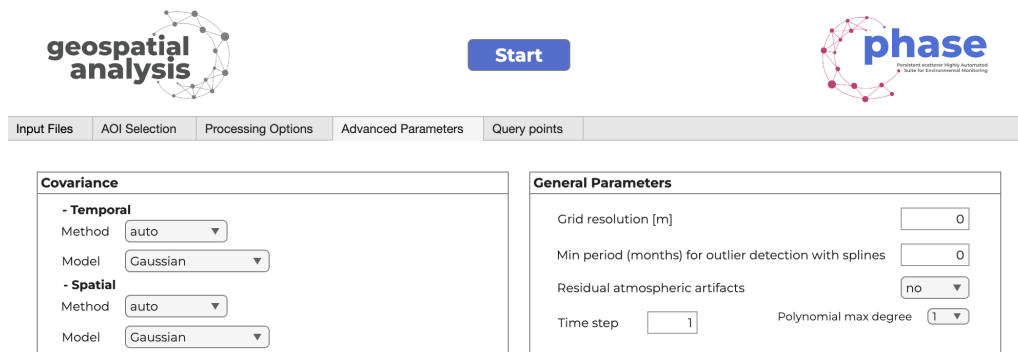
Once projected, the available displacement time series undergo a multi-step processing workflow to filter noise, remove artifacts, and prepare the data for geo-statistical interpolation.

2D spatial domain

When the AOI represents a spatially distributed target—such as an urban center, a volcanic caldera, or a subsidence bowl—constraining the deformation to a 1D centerline is physically inappropriate. To address this, PHASE extends the stochastic modeling framework into a full 2D planar geometry.

The algorithmic pipeline for the 2D case is a direct extension of the 1D workflow. The mathematical estimators, outlier rejection testing, and iterative covariance updates remain fundamentally identical. The core differences lie strictly in the geometric definition of the spatial domain (incorporating the assumption of spatial homogeneity and isotropy) and the 2D expansion of the deterministic trend models.

If this approach is selected, a dedicated graphical interface tab (Figure 4.6) becomes available. The user parameters perfectly mirror the 1D interface, with the key exception being the spatial resolution definition.



The screenshot shows the PHASE software interface. At the top, there's a navigation bar with tabs: 'Input Files', 'AOI Selection', 'Processing Options', 'Advanced Parameters', and 'Query points'. The 'Advanced Parameters' tab is selected. Below the tabs, there are two main panels. The left panel is titled 'Covariance' and contains two sections: '- Temporal' and '- Spatial'. Each section has a 'Method' dropdown set to 'auto' and a 'Model' dropdown set to 'Gaussian'. The right panel is titled 'General Parameters' and contains several input fields: 'Grid resolution [m]' set to 0, 'Min period (months) for outlier detection with splines' set to 0, 'Residual atmospheric artifacts' set to 'no', 'Time step' set to 1, and 'Polynomial max degree' set to 1. Above the panels, there are logos for 'geospatial analysis' and 'phase'.

Figure 4.6: Configuration panel for the 2D stochastic modeling workflow in PHASE. The interface mirrors the 1D setup but adapts the spatial resolution parameter to govern a planar output grid.

In the 2D domain, the **spatial and time resolution** parameter defines a uniform grid spacing applied equally to both the Easting (x) and Northing (y) coordinates, rather than a single curvilinear distance.

WARNING: Similar to the 1D case, displacement values generated over areas lacking PS points are the result of statistical extrapolation. Because 2D extrapolation expands radially across a plane, prediction uncertainties can grow rapidly in large spatial data gaps. Always interpret the results at the edges of your PS point cloud with caution.

4.3.6 Spatio-temporal deterministic (Spline-based)

As an alternative to the stochastic framework, PHASE offers a purely deterministic modeling approach. This methodology abandons empirical covariance estimation and Least Squares Collocation entirely. Instead, the continuous spatio-temporal displacement field is modeled using multidimensional basis splines.

Why choose this mode? Because this approach relies purely on localized geometric fitting rather than global statistical correlation, it does not assume spatial or temporal stationarity. This makes the deterministic approach highly advantageous for modeling highly localized, sudden deformation events (e.g., an isolated structural failure or a sudden sinkhole) that might otherwise be over-smoothed by stochastic covariance averaging. Furthermore, this method is computationally lighter, making it ideal for quick assessments or for executions on machines with lower processing power.

The complete procedural sequence for this deterministic modeling is schematized in Figure 4.7. This logical flowchart applies to both the 1D and 2D spatial domains.

SPATIO-TEMPORAL DETERMINISTIC MODELING

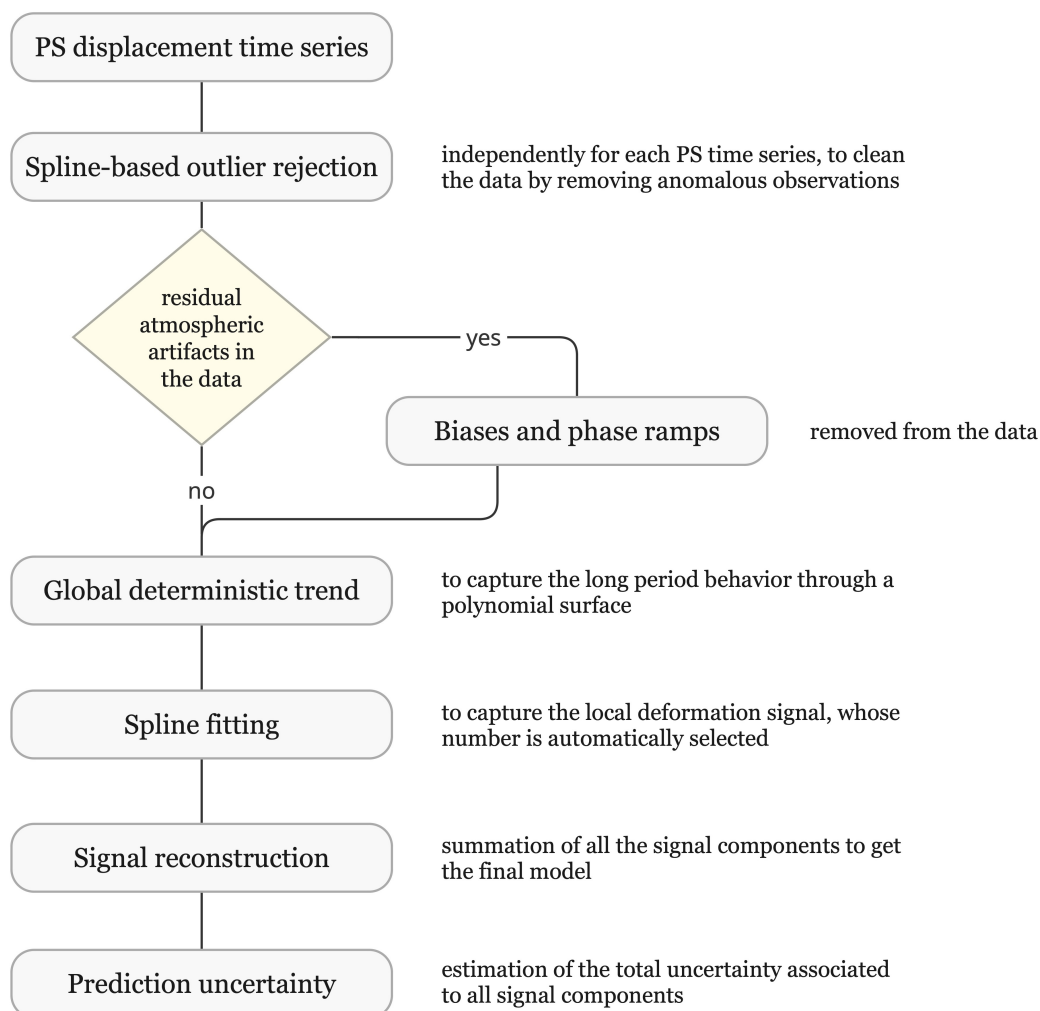


Figure 4.7: Schematic representation of the deterministic spatio-temporal modeling workflow, progressing from raw PS observations through global polynomial trend removal and final multidimensional spline interpolation.

1D spatial domain

Since the 1D infrastructure geometry involves one spatial dimension (ℓ) and one temporal dimension (t), this mode specifically fits a continuous surface using 2D bicubic splines. The entire workflow is configured through a dedicated interface tab (Figure 4.8).

The screenshot displays the PHASE software interface for configuring 1D deterministic modeling. At the top, there is a navigation bar with tabs: 'Input Files', 'AOI Selection', 'Processing Options', 'Advanced Parameters', and 'Query points'. The 'Advanced Parameters' tab is currently selected. Above the tabs, there are three logos: 'geospatial analysis', a blue 'Start' button, and the 'phase' logo. The main configuration area is divided into two panels. The left panel, titled 'Splines', contains three dropdown menus: 'Method' set to 'auto', 'Lambda' set to 'auto', and 'Noise' set to 'auto'. The right panel, titled 'General Parameters', contains several input fields and dropdowns: 'Centerline resolution [m]' with a value of 0, 'Min period (months) for outlier detection with splines' with a value of 0, 'Residual atmospheric artifacts' with a dropdown set to 'no', 'Time step' with a value of 1, and 'Polynomial max degree' with a dropdown set to 1.

Figure 4.8: Configuration panel for the 1D deterministic modeling workflow in PHASE. Users define the spatial/temporal prediction grids, global polynomial trends, and the statistical optimization criteria for the final bicubic spline surface.

Through this interface, you govern the specific estimators within the deterministic framework. Key configurable parameters include:

- **Centerline and time resolution:** defines the uniform spatial ($\Delta\ell$) and temporal (Δt) step sizes for the target prediction grid upon which the final bicubic displacement surface is evaluated.
- **Minimum period for splines fitting:** establishes the maximum flexibility of the 1D splines used during the preliminary temporal outlier rejection step (structurally identical to the stochastic preprocessing).
- **Atmospheric residual errors:** if deemed present, these are estimated and removed. You can specify whether these errors are treated as purely time-variable biases (for small AOIs) or as epoch-specific spatial trends (for large AOIs).
- **Global trend:** defines the degree of a macroscopic spatio-temporal polynomial surface. Unlike the stochastic mode, this global surface is *always* estimated and removed prior to the bicubic spline interpolation to ensure the splines only fit local, high-frequency physical deformations.

- **Bicubic spline optimization:** selects the strict statistical criteria (variance threshold, F-test, MDL or χ^2 test) used to automatically determine the optimal number of spline knots for the final surface, preventing overfitting.

2D spatial domain

When the monitored AOI is a spatially distributed target (e.g., an urban center or a subsidence bowl), this module extends the deterministic, spline-based approach to the planar domain. It reconstructs the continuous deformation field $d(x, y, t)$ using 3D tricubic splines.

Selecting this option activates its corresponding configuration tab (Figure 4.9).

Figure 4.9: Configuration panel for the 2D deterministic modeling workflow in PHASE. The interface mirrors the 1D setup but adapts the spatial resolution and optimization strategies for a full planar grid.

The configuration parameters mirror those of the 1D deterministic case, adapted specifically for the expanded geometric domain. Most notably:

- The **Spatial and time resolution** parameter now defines a uniform square grid spacing applied equally to both the Easting (x) and Northing (y) axes.
- The **Global trend** polynomial degree defines a trivariate surface instead of a bivariate one.

4.3.7 Query points

The filepath to a `.txt` file containing the coordinates (UTM or geographic) of query points can be specified in this last panel. This will trigger the interpolation module of the software to evaluate the estimated spatio-temporal model exactly at these coordinates. In the output Excel report this will result in displacement time series plots in the *Interpolation* sheet.

4.4 Automatic report generation and data export

To facilitate the interpretation and dissemination of the results, the software includes a fully automated workflow for report generation. Upon completion of the processing chain, the software aggregates the numerical results, the estimated statistical parameters, and the generated figures into a comprehensive Microsoft Excel spreadsheet.

The report is structured into five (or six, if external interpolation is requested) specific sheets:

4.4.1 General sheet

This sheet serves as the project summary. It compiles all relevant metadata regarding the analysis, including:

- **Project metadata:** date of execution, municipality, and country of the AOI (automatically geocoded from the reference PS point), and the chosen modeling strategy (e.g., 1D vs. 2D, stochastic vs. deterministic).
- **Input parameters:** details on the PS dataset, including the total number of scatterers, the observation period, and the unwrapping reference point coordinates. If available, the specific StaMPS processing parameters used in the upstream analysis are also reported.
- **Visual context:** three figures are automatically embedded: the project logo, a topographic map locating the AOI within the country, and a satellite map showing the distribution of PS points inside and outside the selected AOI (Figure 4.10).

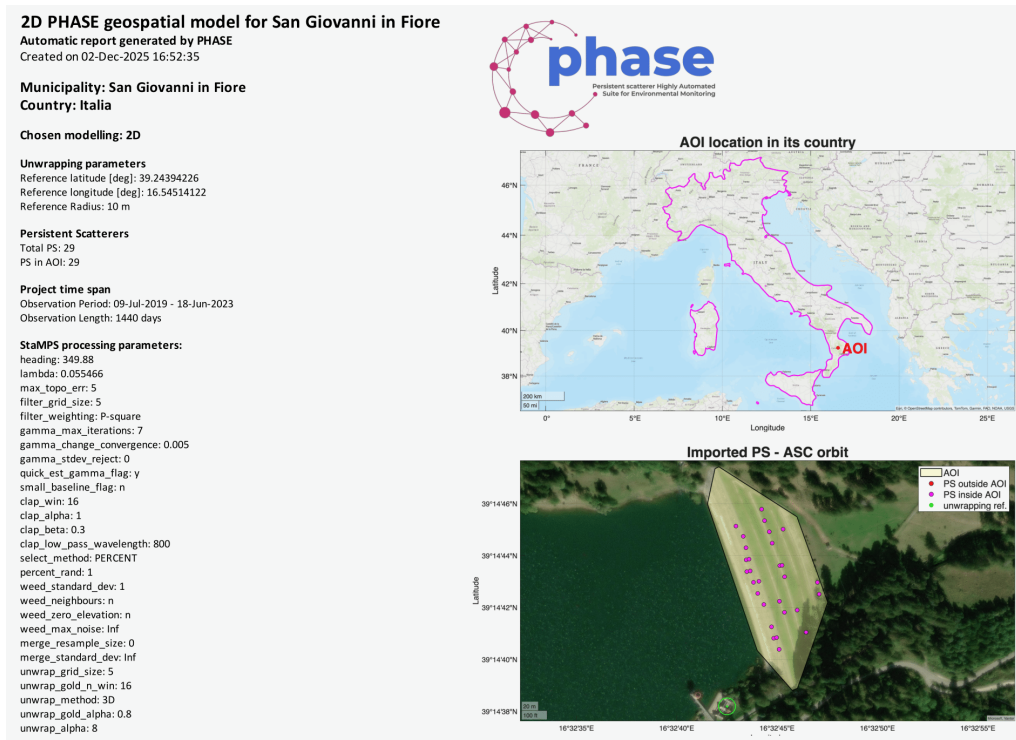


Figure 4.10: Example of the *General* sheet in the automatic output report, showing project metadata and context maps.

4.4.2 Data sheets (Raw, Modeled, Uncertainty)

Following the general overview, the core results are organized into three dedicated sheets. Each shares a similar structure: a header with summary statistics followed by the full data table, where rows represent individual PS points and columns represent observation epochs.

- **Raw displacement:** contains the input PS time series after the initial detrending (subtraction of the modeled value at the first epoch). It includes a plot of the spatially averaged displacement time series over the AOI with an overlaid linear trend to identify macroscopic subsidence or uplift phenomena.
- **Modeled displacement:** contains the final result of the spatio-temporal modeling (stochastic or deterministic). It includes the full set of estimated displacement time series for all positions in the AOI based on the chosen spatial resolution. A map showing the average velocity (in mm/year) is embedded to provide an immediate overview of deformation rates (Figure 4.11).
- **Uncertainty:** mirroring the modeled displacement, this sheet reports the estimated standard deviation for every epoch and point. It includes a map of the time-average uncertainty, allowing you to identify areas where the model is less reliable (e.g., due to low PS density or high noise).

Modelled displacement time series

Each PS displacement modelled time series is obtained through the contribution of a cubic splines deterministic trend and a stochastic component. Coordinates are given in WGS84 (geographic [deg]) and UTM (projected [m] - according to the proper fuse). Displacement is given along the satellite Line Of Sight (LOS) [mm]. The average velocity is in [mm/year].

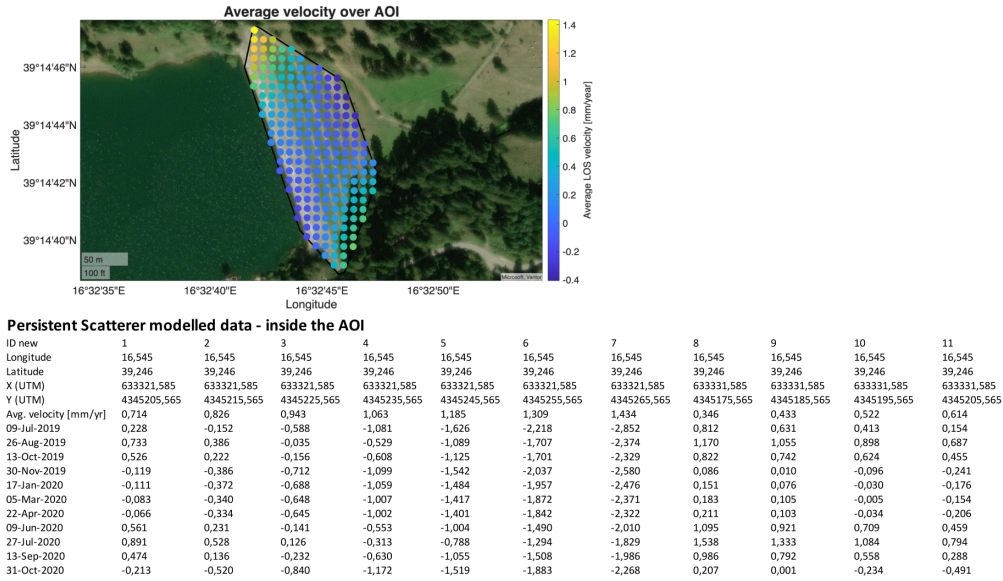


Figure 4.11: Example of the *Modeled displacement* sheet, featuring the data table and the average velocity map.

4.4.3 Interpolation sheet (optional)

If you provide a set of specific target coordinates (e.g., GNSS stations or critical infrastructure points), this sheet reports the displacement time series extracted from the continuous spatio-temporal model at those exact locations. This feature allows you to retrieve the deformation history at arbitrary points, regardless of the original PS distribution. For each target point, a specific plot showing the temporal evolution of the displacement is automatically generated and embedded.

4.4.4 Alerts sheet (risk assessment)

Beyond numerical data, the software provides an *Alerts* sheet featuring a quick risk assessment for every monitored point. The software computes a global score based on two indicators: average velocity and cumulative displacement.

For each point, these metrics are standardized (Z-score normalization) based on the statistics of the entire population. To prioritize only statistically significant deviations, a risk percentage (R_A) is calculated using a formulation that maps the standardized values to probability thresholds:

$$R_A = \begin{cases} 0 & \text{if } z < z_{25} \\ 100 \cdot \left(1 - \exp \left(-a \cdot \frac{z - z_{25}}{z_{2.5} - z_{25}} \right) \right) & \text{if } z_{25} \leq z \leq z_{2.5} \\ 100 + (z - z_{2.5}) \cdot 50 & \text{if } z > z_{2.5}, \end{cases} \quad (4.3)$$

where:

- z is the absolute standardized value of the metric (velocity or cumulative displacement).
- $z_{2.5}$ is the critical value of the normal distribution corresponding to a significance level $\alpha = 5\%$ (two-tailed), representing the threshold for the 95% confidence interval ($z_{2.5} \approx 1.96$).
- z_{25} is the critical value corresponding to a significance level $\alpha = 50\%$ (two-tailed), representing the threshold for the 50% confidence interval ($z_{25} \approx 0.67$).
- a is a sensitivity coefficient (set to 1.5).

This formulation ensures that points falling within the interquartile range of the distribution ($z < z_{25}$) differ negligibly from the mean behavior and are assigned 0% risk. Conversely, points exceeding the 95% confidence threshold ($z > z_{2.5}$) are considered statistical outliers and are assigned a critical risk. The exponential function provides a smooth transition between these two states (Figure 4.12).

Automatic alerts

Alerts thresholds are computed considering both average velocity and cumulative displacement, for each PS. Threshold values are obtained considering a significance level of 5%.

Automatic alerts for PS - inside the AOI

ID new	Longitude	Latitude	X (UTM)	Y (UTM)	Avg velocity [mm/yr]	Velocity risk [%]	Cumulative displ. [mm]	Cumulative displ. risk [%]	Global risk [%]
1	16,545	39,246	633321,585	4345205,565	-0,3	0,0	38,2	5,9	2,9
2	16,545	39,246	633321,585	4345215,565	-0,4	0,0	46,3	0,0	0,0
3	16,545	39,246	633321,585	4345225,565	-0,5	19,7	59,9	0,0	9,9
4	16,545	39,246	633321,585	4345235,565	-0,6	46,4	77,6	0,0	23,2
5	16,545	39,246	633321,585	4345245,565	-0,8	66,6	98,9	57,7	62,1
6	16,545	39,246	633321,585	4345255,565	-1,0	104,3	122,2	113,8	109,1
7	16,545	39,246	633321,585	4345265,565	-1,2	127,3	147,1	157,7	142,5
8	16,545	39,246	633331,585	4345175,565	-0,1	0,0	36,3	13,0	6,5
9	16,545	39,246	633331,585	4345185,565	-0,2	0,0	38,8	3,6	1,8
10	16,545	39,246	633331,585	4345195,565	-0,2	0,0	40,7	0,0	0,0
11	16,545	39,246	633331,585	4345205,565	-0,3	0,0	38,6	4,4	2,2
12	16,545	39,246	633331,585	4345215,565	-0,3	0,0	38,0	6,8	3,4
13	16,545	39,246	633331,585	4345225,565	-0,4	0,0	49,4	0,0	0,0
14	16,545	39,246	633331,585	4345235,565	-0,5	32,3	67,0	0,0	16,1
15	16,545	39,246	633331,585	4345245,565	-0,7	59,1	89,2	37,0	48,0
16	16,545	39,246	633331,585	4345255,565	-0,9	76,0	113,7	77,1	76,5
17	16,545	39,245	633341,585	4345145,565	0,1	0,0	64,3	0,0	0,0
18	16,545	39,245	633341,585	4345155,565	0,0	0,0	57,8	0,0	0,0
19	16,545	39,246	633341,585	4345165,565	-0,1	0,0	46,2	0,0	0,0
20	16,545	39,246	633341,585	4345175,565	-0,1	0,0	40,1	0,0	0,0
21	16,545	39,246	633341,585	4345185,565	-0,2	0,0	47,3	0,0	0,0
22	16,545	39,246	633341,585	4345195,565	-0,2	0,0	49,6	0,0	0,0
23	16,545	39,246	633341,585	4345205,565	-0,2	0,0	43,6	0,0	0,0
24	16,545	39,246	633341,585	4345215,565	-0,2	0,0	36,8	11,4	5,7
25	16,545	39,246	633341,585	4345225,565	-0,3	0,0	36,4	12,6	6,3
26	16,545	39,246	633341,585	4345235,565	-0,4	9,9	53,4	0,0	5,0
27	16,545	39,246	633341,585	4345245,565	-0,6	48,1	77,4	0,0	24,1
28	16,545	39,246	633341,585	4345255,565	-0,9	70,7	103,9	65,7	68,2
29	16,545	39,245	633351,585	4345115,565	0,1	0,0	44,4	0,0	0,0
30	16,545	39,245	633351,585	4345125,565	0,1	0,0	61,3	0,0	0,0
31	16,545	39,245	633351,585	4345135,565	0,1	0,0	71,7	0,0	0,0
32	16,545	39,245	633351,585	4345145,565	0,1	0,0	69,6	0,0	0,0
33	16,545	39,245	633351,585	4345155,565	0,0	0,0	61,2	0,0	0,0
34	16,545	39,246	633351,585	4345165,565	0,0	0,0	49,0	0,0	0,0
35	16,545	39,246	633351,585	4345175,565	-0,1	0,0	37,3	9,5	4,7
36	16,545	39,246	633351,585	4345185,565	-0,2	0,0	42,3	0,0	0,0
37	16,545	39,246	633351,585	4345195,565	-0,1	0,0	44,7	0,0	0,0
38	16,545	39,246	633351,585	4345205,565	-0,1	0,0	42,3	0,0	0,0
39	16,545	39,246	633351,585	4345215,565	0,0	0,0	36,8	11,0	5,5
40	16,545	39,246	633351,585	4345225,565	-0,1	0,0	33,8	21,5	10,7
41	16,545	39,246	633351,585	4345235,565	-0,3	0,0	40,9	0,0	0,0
42	16,545	39,246	633351,585	4345245,565	-0,5	31,0	62,4	0,0	15,5
43	16,545	39,245	633361,585	4345095,565	-0,2	0,0	42,1	0,0	0,0
44	16,545	39,245	633361,585	4345105,565	0,1	0,0	41,1	0,0	0,0
45	16,545	39,245	633361,585	4345115,565	0,2	0,0	49,8	0,0	0,0
46	16,545	39,245	633361,585	4345125,565	0,2	5,3	60,2	0,0	2,6
47	16,545	39,245	633361,585	4345135,565	0,2	0,0	69,6	0,0	0,0
48	16,545	39,245	633361,585	4345145,565	0,1	0,0	68,9	0,0	0,0
49	16,545	39,245	633361,585	4345155,565	0,1	0,0	60,3	0,0	0,0
50	16,545	39,246	633361,585	4345165,565	0,0	0,0	48,6	0,0	0,0
51	16,545	39,246	633361,585	4345175,565	0,0	0,0	35,8	14,8	7,4
52	16,545	39,246	633361,585	4345185,565	0,0	0,0	28,3	37,5	18,8
53	16,545	39,246	633361,585	4345195,565	0,0	0,0	29,5	34,3	17,2
54	16,545	39,246	633361,585	4345205,565	0,1	0,0	34,0	20,9	10,5

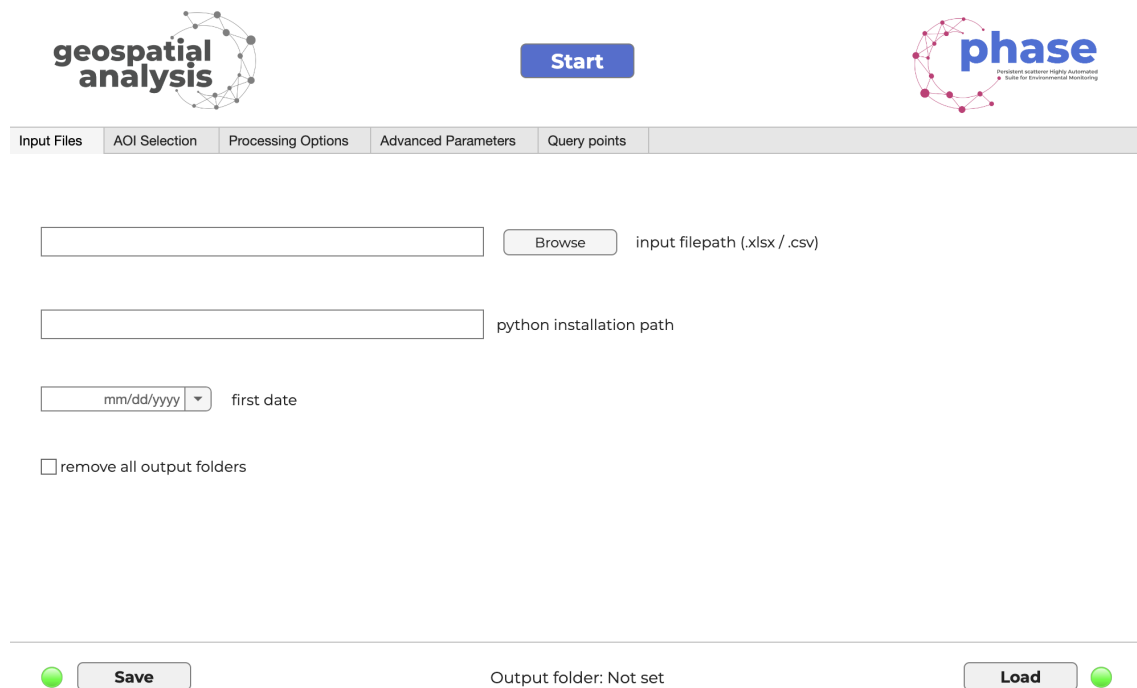
Figure 4.12: Example of the *Alerts* sheet, highlighting points with anomalous velocity or cumulative displacement.

4.4.5 GIS and data export

Finally, in addition to the Excel report, the software automatically exports the results in GIS-compatible formats to facilitate integration with other geospatial workflows:

- **ESRI Shapefile (.shp):** a vector point file containing all points with their attributes (ID, coordinates, average velocity, risk score) and the full displacement time series as additional columns (one for each date). A .prj file is automatically generated to define the Coordinate Reference System (WGS84).
- **MATLAB data (.mat):** a structure array containing the same data for advanced post-processing directly in MATLAB.

These outputs summarize the PSI post-processing, providing ready-to-use data that can be immediately ingested into monitoring algorithms, decision-making chains, and research activities.



The screenshot shows the first panel of the PHASE_model.mlapp application interface. At the top, there is a navigation bar with five tabs: 'Input Files', 'AOI Selection', 'Processing Options', 'Advanced Parameters', and 'Query points'. The 'Input Files' tab is currently selected. Below the navigation bar, the interface is divided into three main sections. On the left is the 'geospatial analysis' logo. In the center is a blue 'Start' button. On the right is the 'phase' logo, which includes the text 'Dispersed, Scalable, Highly Automated Suite for Environmental Monitoring'. Below these sections, there are three input fields. The first is labeled 'input filepath (.xlsx / .csv)' and has a 'Browse' button next to it. The second is labeled 'python installation path'. The third is labeled 'first date' and has a date picker with 'mm/dd/yyyy' format. Below these fields is a checkbox labeled 'remove all output folders'. At the bottom of the interface, there is a 'Save' button on the left, a status indicator 'Output folder: Not set' in the center, and a 'Load' button on the right. The 'Save' and 'Load' buttons are flanked by green circular indicators.

Figure 4.13: PHASE_model.mlapp application interface (first panel).

Appendix



Preprocessing with External DEM

When employing an external Digital Elevation Model (DEM) in the `snap2stamps` procedure, it is mandatory that the file is in the correct format.

A.1 File Requirements

- **File Type:** `.tif` (GeoTIFF)
- **Reference System:** WGS 84 - EPSG:4326

A.2 Preparation Procedure (QGIS)

To prepare the DEM, you can use QGIS to reproject the input DEM according to the source data requirements.

1. Open QGIS and load your DEM.
2. Open the **Warp (Reproject)** tool.
3. Set the **Target CRS** to EPSG:4326.
4. **Important:** Define 0 as the “Nodata value” in the reproject toolbox settings. *(Note: While 0 is not a standard NoData value in general GIS practices, it is strictly required here because the value 0.0 is hardcoded into the snap2stamps XML graph parameters. Using any other value will cause SNAP to crash during processing).*

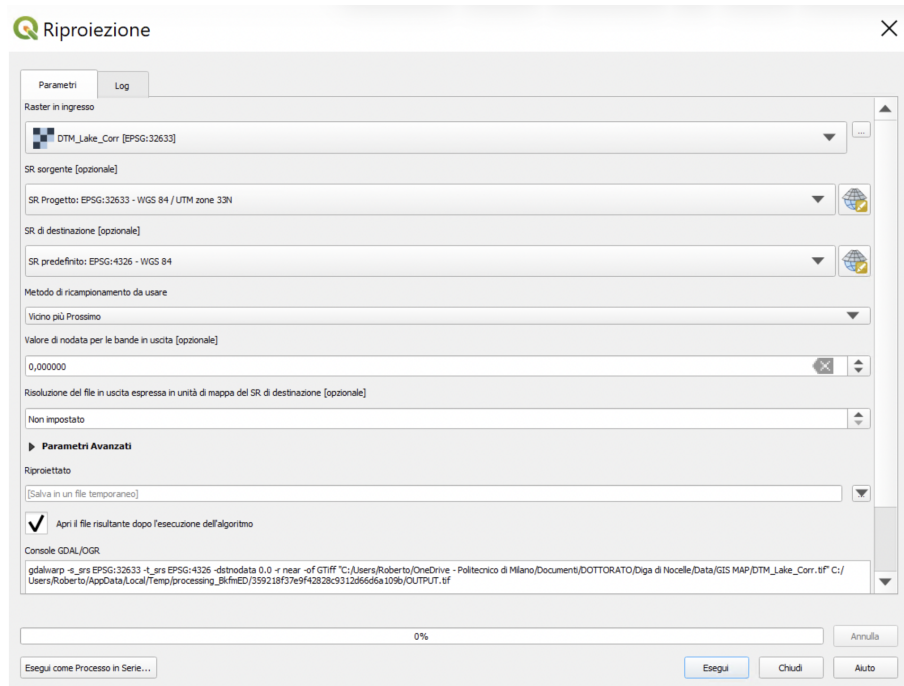


Figure A.1: QGIS Reprojection Settings.

SUGGESTION: For successful processing, it is recommended that the External DEM covers an area that is approximately **10% larger** than the Area of Interest (AOI) set for the subset of the SAR images.